

E-Commerce Sales Analytics Report

Sales Performance Overview

I. SCORECARD SUMMARY

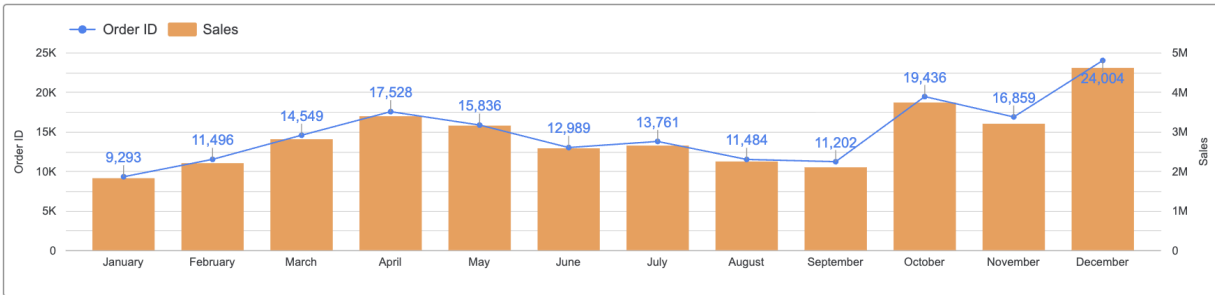
Metric	Value	Detail
Total Revenue	\$34,492,036	The cumulative income generated over the year.
Total Orders	178,437	The total number of unique purchase transactions.
Total Quantity	209,079	The total volume of individual products sold. This number is higher than the Total Orders because a single order may contain multiple products.
Average Order Value (AOV)	\$193.30	The mean value of each transaction.

Scorecard Insights:

- The total quantity (209,079) exceeds the total number of orders (178,437) → some orders may have purchased more than 1 product (avg 1.17 products/order).
- AOV at \$193.30 is relatively high, reflecting a product catalog with many high-value items (Laptops \$1,000–\$1,700, Phones \$400–\$700). This AOV shows that customers are not just buying cheap accessories but are focused on high-priced core products.
- Monthly average: ~\$2.87M revenue and ~14,870 orders. However, actual monthly revenue fluctuates significantly — from \$1.82M (January) to \$4.61M (December), a 2.53x difference — which will be analyzed in detail below.

II. MONTHLY REVENUE TREND — THE BIG PICTURE

Monthly Revenue × Orders



Looking at the Revenue by Month chart overall, revenue in 2019 did NOT grow steadily but instead formed a "skewed W" curve with 3 distinct phases:

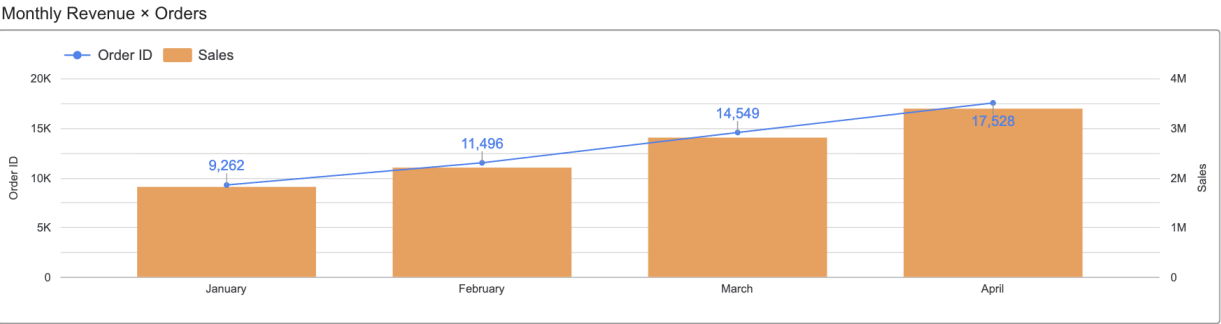
Period	Months	Description	Revenue Range	Change (%)
Growth Phase	Jan → Apr	3 consecutive months up	\$1.82M → \$3.39M	+86.1%
Decline Phase	May → Sep	5-month decline in 2 waves (May-Jun ↓, Jul brief recovery, Aug-Sep ↓)	\$3.15M → \$2.10M	-33.4%
Surge Phase	Oct → Dec	2 peaks (Oct ↑, Nov dip, Dec highest peak)	\$3.74M → \$4.61M	+23.5%

Note: April (Apr) is the peak of the Growth Phase. September (Sep) is the absolute bottom — the pivot point between Decline and Surge. The largest jump of the year occurs from Sep (\$2.10M) → Oct (\$3.74M), an increase of +78.2% in a single month.

The question is: Why do these 3 phases exist? Why does revenue grow then decline? Why

does it explode after declining? Let's dive deep into each phase.

III. GROWTH PHASE (JANUARY → APRIL): Post-holiday recovery and steady demand buildup



Metric	January	February	March	April
Revenue	\$1,822,257 ▾	\$2,202,022	\$2,807,100	\$3,390,670
MoM Revenue Change	— ▾	+20.8%	+27.5%	+20.8%
Orders	9,293 ▾	11,496	14,549	17,528
MoM Order Change	— ▾	+23.7%	+26.6%	+20.5%

Over 3 consecutive months, revenue grew at an average of +23.0%/month. Total increase from Jan to Apr was +86.1% (+\$1,568,414).

3.1 JANUARY PROFILE — What does the lowest month look like?

January recorded just \$1.81M with 9,262 orders — the lowest of all 12 months. Breaking down by product:

SubCategory	Revenue Jan	% of Revenue	Orders	Avg Price
Laptops	\$614,798	33.9%	450	\$1,366
Phones	\$506,100	27.9%	817	\$617
Monitors	\$327,487	18.1%	1,255	\$265
Headphones	\$200,955	11.1%	2,457	\$82

TVs	\$72,900	4.0%	243	\$300
Laundry	\$48,600	2.7%	81	\$600
Cables	\$32,509	1.8%	2,227	\$13
Batteries	\$10,237	0.6%	2,114	\$3

Key observations:

- HIGH-VALUE items (Laptops \$1,366 + Phones \$617) account for 61.8% of revenue but only 1,267 orders (13.1% of total) — spending on expensive products is at a very low level
- LOW-PRICE items (Cables \$13 + Batteries \$3) dominate order count (4,341 orders = 45.0%) but contribute only 2.4% of revenue
- Headphones (2,457 orders) sits in between: high volume but at \$82/unit, moderate revenue contribution (11.1%)
- Laundry + TVs: very few orders (81 + 243) — these are bulky/high-value products with the lowest purchase volume

→ January shows a clear pattern: buyers still place orders (mostly small, cheap accessories), but HIGH-VALUE PRODUCTS are purchased at very low volume — notably Laptops at just 450 orders, Laundry at only 81 orders.

3.2 FEBRUARY-APRIL: DEMAND RECOVERS — WHICH GROUPS GREW?

SubCategory	Jan → Apr Revenue	Growth %	Δ Absolute
Laptops	\$615K → \$1,162K	+89.0%	+\$546,998
Phones	\$506K → \$921K	+82.0%	+\$415,000
Monitors	\$327K → \$623K	+90.2%	+\$295,379
Headphones	\$201K → \$381K	+89.7%	+\$180,342
TVs	\$73K → \$138K	+89.3%	+\$65,100
Laundry	\$49K → \$83K	+71.6%	+\$34,800
Cables	\$33K → \$63K	+92.5%	+\$30,081
Batteries	\$10K → \$20K	+91.7%	+\$9,383

Key findings:

- ALL 8 SubCategories grew +71% to +92% — no group was left behind

- However, ABSOLUTE increases differ dramatically: Top 3 (Laptops + Phones + Monitors) contributed +\$1,257,377, accounting for 79.7% of the total \$1,577,084 increase
- Laundry grew the slowest (+71.6%) and was the ONLY group that declined from Jan→Feb (-3.7%) before recovering sharply Feb→Mar (+44.9%)

3.3 THE FIXED REVENUE ENGINE — Does the product mix shift between low and high months?

Revenue share comparison between the lowest month (Jan) and highest month (Apr):

SubCategory	% Jan	% Apr	Change
Laptops	33.9%	34.3%	+0.3%
Phones	27.9%	27.2%	-0.7%
Monitors	18.0%	18.4%	+0.3%
Headphones	11.1%	11.2%	+0.2%
TVs	4.0%	4.1%	+0.1%
Laundry	2.7%	2.5%	-0.2%
Cables	1.8%	1.8%	+0.1%
Batteries	0.6%	0.6%	0.0%

Remarkable finding: the revenue share DOES NOT CHANGE (maximum shift of only 0.7pp).
This means:

- Revenue structure is extremely stable — even though April revenue is 1.86x January, each product group's contribution remains nearly identical
- Laptops + Phones + Monitors consistently account for ~80% of revenue in BOTH the lowest and highest months → this is the FIXED GROUP that determines the business's revenue
- No "shift" occurred — buyers did not switch from expensive to cheap products or vice versa; ALL GROUPS grew at the same pace

→ The Growth Phase is a classic "rising tide lifts all boats" pattern: the entire market grew uniformly, with no single product group driving a breakout or falling behind. The only difference between the lowest and highest months is ORDER VOLUME, not product mix.

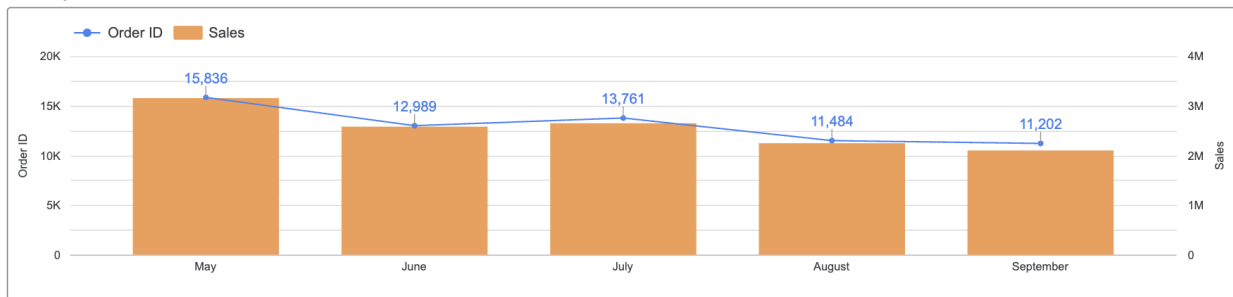
3.4 CUSTOMER BEHAVIOR CONTEXT

January falls right after the biggest shopping season of the year (Black Friday, Christmas, New Year) — a period when most purchasing demand was already released in November-December. Customers enter January in a "already bought enough" state, pushing order volume to the year's lowest (9,262 orders).

From February onward, purchasing demand recovered naturally following the consumption cycle: electronics have usage and replacement lifecycles, accessories need periodic replenishment. Notably, the recovery pace was very consistent (+21-27%/month) and spread across ALL product groups — indicating this was not triggered by any specific event, but rather the market's natural return to baseline after a post-holiday spending pause.

IV. DECLINE PHASE (MAY → SEPTEMBER): 5 months of decline in 2 waves — which product groups were hit hardest?

Monthly Revenue × Orders



Month	Revenue	MoM Change	Orders	MoM Orders
May	\$3,152,607	-7.0%	15,836	-9.7%
Jun	\$2,577,802	-18.2%	12,989	-18.0%
Jul	\$2,647,776	+2.7%	13,761	+5.9%
Aug	\$2,244,468	-15.2%	11,484	-16.6%
Sep	\$2,097,560	-6.5%	11,202	-2.5%

Right after the April peak, revenue declined continuously for 5 months (except a brief recovery in July). Total decline from May to Sep: -33.4% (-\$1,055,047). If measured from the Apr peak: -38.1% (-\$1,293,110).

4.1 WAVE 1 (APR → JUN): Not all categories declined at the same pace

The transition from Apr (peak) to May reveals an interesting pattern — the decline did NOT hit all product groups equally:

SubCategory	Revenue Apr	Revenue May	Apr→May	Revenue Jun	May→Jun
Laptops	\$1,161,796	\$1,161,496	-0.03%	\$919,197	-20.9%
Laundry	\$83,400	\$84,000	+0.7%	\$58,800	-30.0%
Phones	\$921,100	\$810,000	-12.1%	\$669,700	-17.3%
Monitors	\$622,867	\$558,939	-10.3%	\$469,242	-16.0%
Headphones	\$381,296	\$344,831	-9.6%	\$291,013	-15.6%
TVs	\$138,000	\$119,700	-13.3%	\$110,100	-8.0%
Cables	\$62,591	\$55,734	-11.0%	\$45,237	-18.8%
Batteries	\$19,620	\$17,906	-8.7%	\$14,513	-19.0%

Key observation:

- Laptops was virtually FLAT (-0.03%, just \$300 difference) — the highest-priced product showed the strongest resistance to decline
- Laundry actually INCREASED (+0.7%) — another high-value product holding steady
- Meanwhile Phones (-12.1%), TVs (-13.3%), and Monitors (-10.3%) dropped immediately
- This suggests: buyers of the most expensive items (Laptops \$1,366, Laundry \$600) still purchased in May, while mid-range and accessory categories pulled back first

By June, the decline deepened across ALL categories. The sharpest drops: Laundry -30.0% (the May holdout collapsed), Laptops -20.9% (finally fell hard), Batteries -19.0%, Cables -18.8%.

4.2 JULY RECOVERY: Which categories bounced back?

July showed a brief +2.7% recovery. But it was NOT uniform:

SubCategory	Jun → Jul	Recovery pattern
Laundry	+9.2% ▾	Strongest bounce ▾
TVs	+8.7% ▾	Strong bounce ▾
Cables	+8.7% ▾	Strong bounce ▾
Headphones	+5.8% ▾	Moderate ▾
Monitors	+4.6% ▾	Moderate ▾

Batteries	+3.5% ▾	Weak ▾
Laptops	+2.8% ▾	Weak ▾
Phones	-2.0% ▾	NO recovery ▾

Notable: Phones was the ONLY category that did NOT recover in July — it continued declining (-2.0%). This makes Phones the most consistently declining product across the entire phase: it dropped every single month from Apr to Sep without exception.

4.3 WAVE 2 (AUG → SEP): Continued decline to the bottom

After July's brief bounce, Aug-Sep resumed the downward trend. By September (the absolute bottom):

SubCategory	Apr → Sep Revenue	Decline %	Revenue Change
Laundry	\$83K → \$46K	-45.3% ▾	-\$37,800
Phones	\$921K → \$533K	-42.1% ▾	-\$388,200
Monitors	\$623K → \$396K	-36.5% ▾	-\$227,352
Laptops	\$1,162K → \$739K	-36.4% ▾	-\$423,199
Headphones	\$381K → \$244K	-36.1% ▾	-\$137,556
TVs	\$138K → \$88K	-36.1% ▾	-\$49,800
Batteries	\$20K → \$13K	-36.0% ▾	-\$7,070
Cables	\$63K → \$40K	-35.4% ▾	-\$22,134

Key findings:

- Laundry declined the MOST (-45.3%) — from being the last holdout in May to the steepest faller overall
- Phones declined second most (-42.1%) and was the only product that declined EVERY month without exception
- The remaining 6 categories clustered tightly at -35% to -36.5%, showing a very uniform decline
- Top 3 (Laptops + Phones + Monitors) contributed -\$1,038,750, accounting for 80.3% of the total decline — mirroring the 79.7% contribution during growth

4.4 REVENUE STRUCTURE: Did the product mix shift during the decline?

SubCategory	% Apr	% Sep	Change
Laptops	34.3%	35.2%	+0.9% ▾
Phones	27.2%	25.4%	-1.8% ▾
Monitors	18.4%	18.9%	+0.5% ▾
Headphones	11.2%	11.6%	+0.4% ▾
TVs	4.1%	4.2%	+0.1% ▾
Laundry	2.5%	2.2%	-0.3% ▾
Cables	1.8%	1.9%	+0.1% ▾
Batteries	0.6%	0.6%	0.0% ▾

The only notable shift: Phones lost -1.8% share (the steepest consistent decliner), while Laptops gained +0.9% (held up longest in May). But overall, the structure remained remarkably stable — Laptops + Phones + Monitors still accounted for 79.4% at the bottom vs 79.8% at the peak.

→ The Decline Phase mirrors the Growth Phase in reverse: order volume dropped while the product mix stayed nearly unchanged. The key difference is in the TIMING — not all categories declined simultaneously. Laptops and Laundry resisted initially (May), then fell sharply (Jun onward). Phones was the most vulnerable, declining every single month. But by September, all categories had converged to a similar ~36% decline, confirming the same "falling tide lowers all boats" pattern.

4.5 CUSTOMER BEHAVIOR CONTEXT

The May-Sep period coincides with the US summer — a time when consumer behavior shifts distinctly: people prioritize travel, outdoor activities, and experience-based spending over online product purchases. This explains why order volume declined evenly across ALL product groups, not just specific categories.

July's brief recovery (+2.7%) coincides with mid-year shopping events in the US (July 4th Independence Day sales, Amazon Prime Day) — creating a short-term demand stimulus before the downward trend continued.

September hit the deepest bottom not just because of extended summer effects, but also due to the "anticipation effect": customers know the biggest sale season is approaching (Black Friday,

Holiday Season) and DELIBERATELY delay purchases. This is a well-documented e-commerce phenomenon called the "pre-holiday spending freeze" — Sep AOV also dropped to \$187.25 (the year's lowest), showing that even those who did buy shifted toward cheaper items.

5. COMPARING GROWTH vs DECLINE: WHAT'S DIFFERENT? WHY?

Metric	Growth Phase (Jan→Apr)	Decline Phase (May→Sep)
Duration	3 months	5 months
Amplitude of change	+86.1% (+\$1.57M)	-33.4% (-\$1.06M)
Speed	+23.0%/month	-6.7%/month
Driver	Volume surging	Volume gradually fading
AOV	Stable (\$191–\$196)	Stable then declining (\$187–\$199)
Category mix	Uniform across all	Uniform across all

The most important difference: **SPEED**

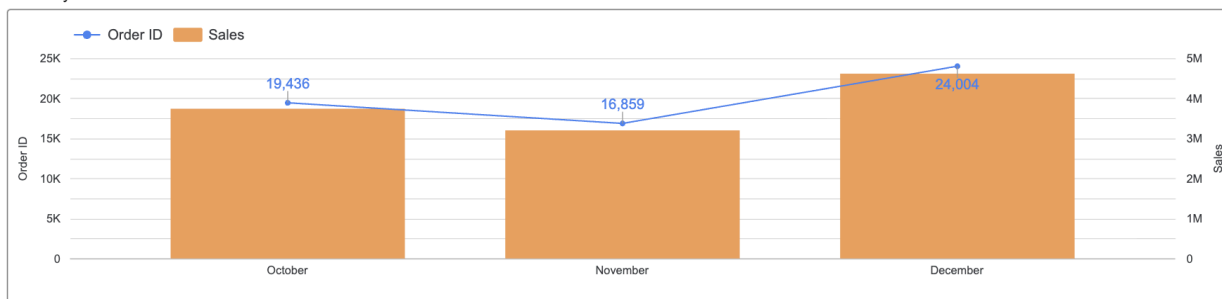
- The growth phase was MUCH FASTER: +23%/month over 3 months
- The decline phase was SLOWER: -6.7%/month but stretched over 5 months
- Meaning: Customers return to shopping QUICKLY after the holiday break, but leave SLOWLY during summer — a positive signal for retention

Similarities:

- Both phases were driven by VOLUME (order count), not VALUE (AOV)
- Category mix did not change — Electronics consistently held 83-84%, product structure remained stable
- This confirms: revenue fluctuations are entirely due to CUSTOMER COUNT, not changes in shopping behavior

6. SURGE PHASE (OCTOBER → DECEMBER): Explosion from the bottom with 2 peaks — which product groups led the recovery?

Monthly Revenue × Orders



After hitting bottom at September (\$2.10M — the pivot point), revenue exploded back:

Month	Revenue	MoM Change	Orders	MoM Orders
Oct	\$3,736,727	+78.2%*	19,436	+73.5%*
Nov	\$3,199,603	-14.4%	16,859	-13.3%
Dec	\$4,613,443	+44.2%	24,004	+42.4%

(*MoM compared to Sep — the pivot point bottom month)

This is the most impressive phase: from the Sep bottom, revenue increased +119.9% in just 3 months to the \$4.61M peak (Dec). Within the Surge Phase itself (Oct→Dec), revenue grew +23.5% from \$3.74M to \$4.61M.

(a) PEAK 1 — OCTOBER: The largest leap of the year (+78.1%)

Sep → Oct nearly DOUBLED in a single month. Breaking down by product group:

SubCategory	Revenue Sep	Revenue Oct	Sep→Oct
Laptops	\$738,598	\$1,353,495	+83.3%
TVs	\$88,200	\$157,200	+78.2%
Monitors	\$395,515	\$702,764	+77.7%
Phones	\$532,900	\$940,600	+76.5%
Headphones	\$243,741	\$424,401	+74.1%
Cables	\$40,457	\$70,322	+73.8%
Batteries	\$12,550	\$21,345	+70.1%

Laundry	\$45,600	\$66,600	+46.1%
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Key findings:

- 7 out of 8 groups surged +70% to +83% — very uniform growth, mirroring the "rising tide" pattern from the Growth Phase
- Laptops led the recovery (+83.3%) — the group that held out longest during the Decline Phase now also recovered the strongest
- Laundry is the SOLE OUTLIER: only +46.1%, far below the other groups. This was the steepest decliner in the Decline Phase (-45.3%) and now also the slowest to recover — indicating this group suffered deeper demand damage and needs more time to bounce back

(b) NOVEMBER DIP: All groups declined, but at very different rates

November dropped -14.4% vs October — a dip between the two peaks. Looking at the details:

SubCategory	Revenue Oct	Revenue Nov	Oct→Nov
Laptops	\$1,353,495	\$1,121,996	-17.1%
Monitors	\$702,764	\$587,657	-16.4%
Cables	\$70,322	\$59,999	-14.7%
TVs	\$157,200	\$138,000	-12.2%
Headphones	\$424,401	\$375,326	-11.6%
Phones	\$940,600	\$832,100	-11.5%
Batteries	\$21,345	\$19,125	-10.4%
Laundry	\$66,600	\$65,400	-1.8%

Key findings:

- Laptops declined the MOST (-17.1%) — the opposite of its behavior in the Decline Phase, where it held out the longest. This shows October absorbed a massive wave of Laptop demand, causing a sharp November pullback
- Monitors (-16.4%) also dropped steeply — high-priced Electronics were hit hardest
- Laundry barely declined (-1.8%) — the slowest recoverer in October now held the steadiest in November, as its demand hadn't been fully released yet

(c) PEAK 2 — DECEMBER: Highest peak of the year (\$4.61M)

Nov → Dec surged +44.2%, reaching a record high. December contributed 13.4% of total annual revenue — this single month equals 2.53x January.

SubCategory	Revenue Nov	Revenue Dec	Nov→Dec	Revenue Change
Laptops	\$1,121,996	\$1,632,095	+45.5%	+\$510,098
Phones	\$832,100	\$1,176,800	+41.4%	+\$344,700
Monitors	\$587,657	\$866,438	+47.4%	+\$278,780
Headphones	\$375,326	\$526,742	+40.3%	+\$151,416
Laundry	\$65,400	\$99,600	+52.3%	+\$34,200
TVs	\$138,000	\$199,800	+44.8%	+\$61,800
Cables	\$59,999	\$84,988	+41.6%	+\$24,989
Batteries	\$19,125	\$26,981	+41.1%	+\$7,856

Key findings:

- Laundry surged the MOST (+52.3%) — a delayed explosion after "lagging" in October and holding steady in November
- Monitors (+47.4%) and Laptops (+45.5%) also bounced back strongly after being the deepest decliners in November
- All groups grew +40% to +52% — uniform growth confirming the broad-based recovery pattern

d) FULL SURGE: FROM BOTTOM (SEP) TO PEAK (DEC) — Did the product mix shift?

SubCategory	Sep → Dec Revenue	Growth %	Revenue Change
TVs	\$88K → \$200K	+126.5%	+\$111,600
Laptops	\$739K → \$1,632K	+121.0%	+\$893,497
Phones	\$533K → \$1,177K	+120.8%	+\$643,900
Monitors	\$396K → \$866K	+119.1%	+\$470,923
Laundry	\$46K → \$100K	+118.4%	+\$54,000

Headphones	\$244K → \$527K	+116.1%	+\$283,002
Batteries	\$13K → \$27K	+115.0%	+\$14,430
Cables	\$40K → \$85K	+110.1%	+\$44,531

Total increase: +\$2,515,883. Top 3 (Laptops + Phones + Monitors) contributed +\$2,008,320 = 79.8% — NEARLY IDENTICAL to the contribution during growth (79.7%) and decline (80.3%).

Revenue share:

SubCategory	% Sep	% Dec	Change
Laptops	35.2%	35.4%	+0.2% ▾
Phones	25.4%	25.5%	+0.1% ▾
Monitors	18.9%	18.8%	-0.1% ▾
Headphones	11.6%	11.4%	-0.2% ▾
TVs	4.2%	4.3%	+0.1% ▾
Laundry	2.2%	2.2%	0.0% ▾
Cables	1.9%	1.8%	-0.1% ▾
Batteries	0.6%	0.6%	0.0% ▾

→ The Surge Phase confirms for the third time the same pattern: revenue surged +119.9% yet each product group's share barely moved (maximum 0.2% shift). Whether at the bottom or the peak, Laptops + Phones + Monitors consistently account for ~80% of revenue. The key difference from Growth and Decline: within the Surge Phase, there is a PHASE LAG between groups — Laundry recovered one beat slower (weak Oct, steady Nov, burst in Dec), while Laptops and Monitors surged hardest in Oct then dipped deepest in Nov before reaching the Dec peak.

(e) CUSTOMER BEHAVIOR CONTEXT

The Surge Phase coincides with the largest year-end shopping season in the US — a chain of consecutive events that strongly stimulate consumer spending:

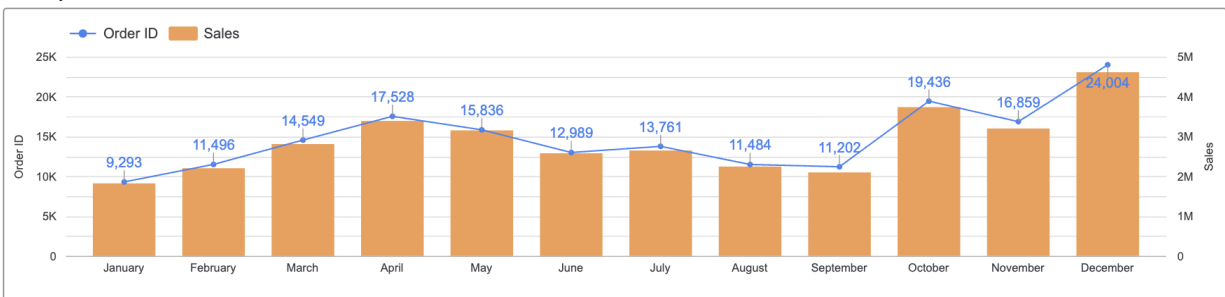
- OCTOBER: The holiday shopping season begins early. After 5 months of decline (May-Sep), pent-up demand is released all at once. Early holiday deals and pre-Black

Friday promotions appear from early October, creating the +78.1% jump. The data shows ALL groups (except Laundry) surged +70-83%, confirming this was broad-based purchasing demand, not driven by any single event.

- **NOVEMBER:** Despite Black Friday (end of month) and Cyber Monday, revenue actually DROPPED -14.4% vs October. Behavioral explanation: October had already absorbed a large portion of early demand. During the first 3 weeks of November, customers shifted into a "waiting for the exact sale day" mode → revenue concentrated mostly in the final week but couldn't compensate for 3 quiet weeks. Additionally, some Black Friday orders may have fallen into early December due to order processing times.
- **DECEMBER:** Christmas shopping season — the biggest gift-buying period of the year. This is why December surpassed even October: gift-buying demand stimulated purchases across all categories, especially Laptops and Phones — the two most popular gift products, contributing +\$510K and +\$345K in absolute growth from Nov→Dec.

7. COMPARING THE 3 PEAK MONTHS: OCT vs NOV vs DEC

Monthly Revenue × Orders



Metric	Oct	Nov	Dec
Revenue	\$3,736,727	\$3,199,603	\$4,613,443
Orders	19,436	16,859	24,004
AOV	\$192.26	\$189.79	\$192.19
Annual Share	10.8%	9.3%	13.4%

These 3 months combined = \$11,549,773 = 33.5% of total annual revenue. Meaning 1/4 of the time (3/12 months) generates 1/3 of revenue.

Why is it Dec > Oct > Nov (not Dec > Nov > Oct)?

- Oct: The "spring rebound" after 5 months of decline — pent-up demand released
- Nov: Customers waiting for Black Friday → revenue "stuck" at end of month, couldn't

compensate

- Dec: Christmas gift-buying pressure + year-end deals → everyone spends, including those who already bought in Oct

Interesting point: AOV barely CHANGED across the 3 months (\$189–\$192)

- The holiday season doesn't make customers buy more expensive items — they just BUY MORE
- Dec had 24,004 orders vs Oct 19,436 → a difference of 4,568 orders, which is exactly the additional "gift buyers"

8. REVENUE AND ORDERS CORRELATION: A VOLUME-DRIVEN BUSINESS

Month	Revenue	Orders	AOV
Jan	\$1,822,257	9,293	\$196.09
Feb	\$2,202,022	11,496	\$191.55
Mar	\$2,807,100	14,549	\$192.94
Apr	\$3,390,670	17,528	\$193.44
May	\$3,152,607	15,836	\$199.08
Jun	\$2,577,802	12,989	\$198.46
Jul	\$2,647,776	13,761	\$192.41
Aug	\$2,244,468	11,484	\$195.44
Sep	\$2,097,560	11,202	\$187.25
Oct	\$3,736,727	19,436	\$192.26
Nov	\$3,199,603	16,859	\$189.79
Dec	\$4,613,443	24,004	\$192.19

AOV fluctuates within an extremely narrow range: \$187.25 (Sep) → \$199.08 (May), a spread of only \$11.83.

Meanwhile, Orders fluctuate widely: 9,293 (Jan) → 24,004 (Dec), a 2.58x spread.

Conclusion: This is a 100% Volume-Driven Business. Revenue increases and decreases are entirely dependent on ORDER COUNT. The value per order ($\$193 \pm \6) is virtually a constant. All revenue growth strategies must focus on ACQUISITION (attracting more customers) and

FREQUENCY (increasing purchase frequency), not UPSELL (increasing order value).

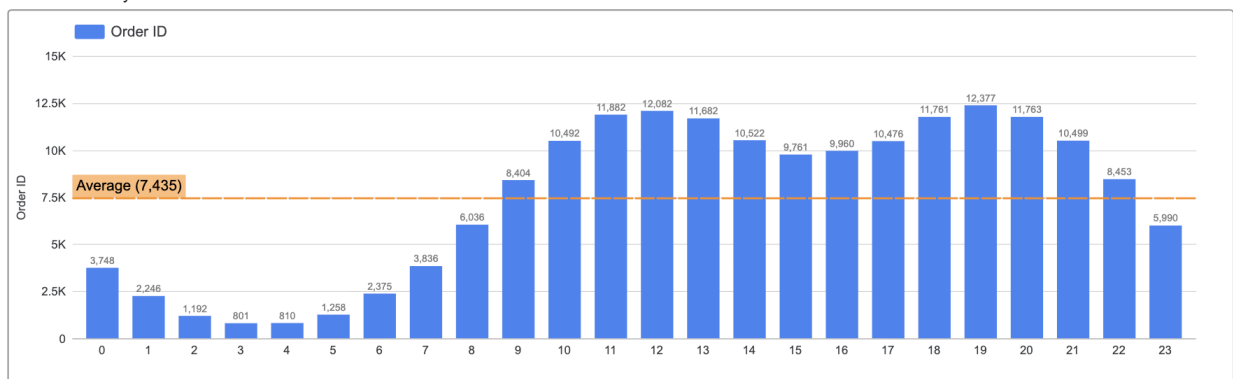
9. KEY TAKEAWAYS

- **REVENUE FOLLOWS A 3-PHASE CYCLE:** Growth (Jan-Apr) → Decline (May-Sep) → Surge (Oct-Dec). This cycle accurately reflects US e-commerce consumer rhythms: post-holiday recovery → summer lull → holiday explosion.
- **VOLUME DETERMINES EVERYTHING:** AOV is stable at $\$193 \pm \6 across all 12 months. All revenue fluctuations are driven by order count. Strategy must prioritize acquiring new customers and increasing purchase frequency.
- **Q4 IS "HARVEST SEASON":** The final 3 months contribute 33.5% of annual revenue. Marketing budget should be heavily allocated to Q4, especially October (the spring rebound) and December (the absolute peak).
- **SEPTEMBER IS AN OPPORTUNITY:** The deepest bottom (\$2.10M) right before the explosion. An "early bird deals" campaign in September could pull some Q4 demand forward, flattening the bottom and reducing December logistics pressure.
- **DEC/JAN RATIO = 2.53x:** The gap between the highest and lowest months is too large. A customer retention strategy post-holiday season is needed to reduce the "post-holiday slump" in January.

PURCHASING TIME PATTERNS — INSIGHT ANALYSIS

1. THE BIG PICTURE — "TWIN PEAK" MODEL

Total Orders by Hour



Looking at the Total Orders by Hour chart overall, online shopping behavior throughout the day forms a clear "Twin Peak" pattern:

Metric	Value
Average Line	7,434 orders/hour
Hours above average	14 hours (9:00–22:00)
Hours below average	10 hours (23:00–8:00)
Highest hour	19:00 (12,377 orders)
Lowest hour	3:00 (801 orders)
Peak/Trough ratio	15.5x

Two peaks appear at:

- Lunch peak: 12:00 (12,082 orders)
- Evening peak: 19:00 (12,377 orders) — higher than the lunch peak

The question: Why are there 2 peaks? Why is the evening peak higher than the lunch peak? Let's dive deep into each time slot.

2. E-COMMERCE BEHAVIORAL SEGMENTS BY TIME

Online shopping behavior follows customers' daily life rhythms. Unlike brick-and-mortar stores (limited by opening hours), e-commerce operates 24/7 — but customers still shop according to their "life rhythm." This can be divided into 7 behavioral segments:

Segment	Time Slot	Orders	Share	E-commerce Behavior
Deep Sleep	2:00–5:00	4,061	2.3%	Nearly zero online activity
Wake & Browse	6:00–8:00	12,247	6.9%	Waking up, checking phone in bed, quick browsing
Commute & Work	9:00–11:00	30,778	17.2%	Browsing on commute, tab shopping

				between work tasks
Lunch Break	12:00–13:00	23,764	13.3%	Office lunch break, concentrated shopping
Afternoon Plateau	14:00–17:00	40,719	22.8%	Shopping between work tasks, slight decline
Evening at Home	18:00–21:00	46,400	26.0%	Home on couch/computer, intentional shopping
Late Night Wind-Down	22:00–1:00	20,437	11.5%	Phone browsing in bed, gradually decreasing

3. DEEP DIVE INTO EACH TIME SLOT — WHAT DO CUSTOMERS BUY, AND WHEN?

Before diving into each time slot, here's an overview of product mix and AOV across all 7 segments:

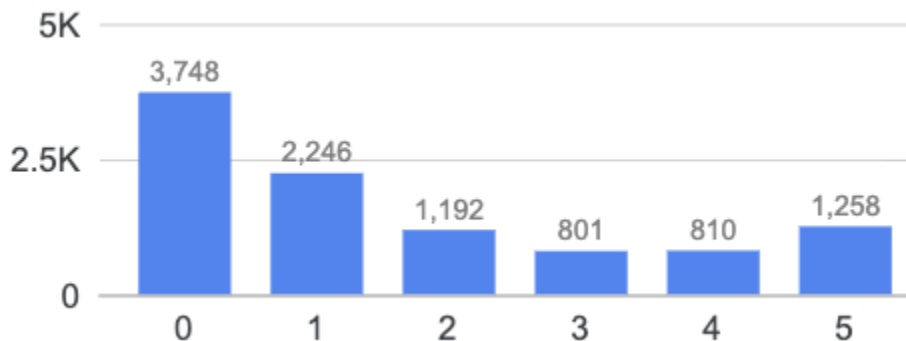
Segment	AOV	Laptops	Phones	Monitors	Headphones	Others
Deep Sleep (2-5h)	\$189.67	34.5% ▾	25.9% ▾	19.1%	11.5%	9.0% ▾
Wake & Browse (6-8h)	\$194.77	36.4% ▾	25.8% ▾	17.7%	11.2%	8.9% ▾
Commute & Work (9-11h)	\$191.17	34.8% ▾	26.5% ▾	18.3%	11.7%	8.7% ▾

Lunch Break (12-13h)	\$188.19	34.1% ▾	26.0% ▾	18.9%	11.9%	9.1% ▾
Afternoon (14-17h)	\$197.92	36.4% ▾	25.2% ▾	18.4%	11.0%	9.0% ▾
Evening (18-21h)	\$193.02	35.0% ▾	26.0% ▾	18.6%	11.4%	9.0% ▾
Late Night (22-1h)	\$193.59	35.1% ▾	26.1% ▾	18.7%	11.6%	8.5% ▾
Overall average	\$193.29	~35% ▾	~26% ▾	~18.5%	~11.5%	~9% ▾

KEY FINDING: Product mix does NOT change by time of day. Laptops always account for ~35%, Phones ~26%, Monitors ~18.5% — whether it's 3 AM or 7 PM. AOV fluctuates within a narrow range of \$188–\$198, with a maximum gap of only \$9.73.

→ This means: time of day changes HOW MANY people buy, but does NOT change WHAT THEY BUY.

3.1. DEEP SLEEP — 2:00–5:00 (2.3%, 4,061 orders, AOV \$189.67)



- 4 hours produce only 4,061 orders and \$770,245 revenue — LESS than a single peak hour (19:00 = 12,377 orders)
- Absolute trough: 3:00 AM — 801 orders
- AOV \$189.67 — slightly below average, but product mix remains identical (Laptops 34.5%, Phones 25.9%)

This is physiological sleep time — virtually no one is awake to shop. Yet e-commerce still records 801 orders at 3AM: there's always a small group of "night owls" or customers in other

time zones placing orders. A brick-and-mortar store would have 0 orders during this window — this is a natural advantage of the online channel.

3.2. WAKE & BROWSE — 6:00–8:00 (6.9%, 12,247 orders, AOV \$194.77)

- Orders ramp up fastest of the day: 6:00 (2,375) → 7:00 (3,836) → 8:00 (6,036) — up 154% in just 2 hours
- Revenue: \$2,385,316 (6.9% of total)
- AOV \$194.77 — above average, product mix stable (Laptops 36.4%)

Characteristic smartphone behavior: wake up → check phone in bed → see products saved from the night before → place a quick order before leaving home. Many orders in this window are decisions "incubated" overnight.

3.3. COMMUTE & WORK — 9:00–11:00 (17.2%, 30,778 orders, AOV \$191.17)

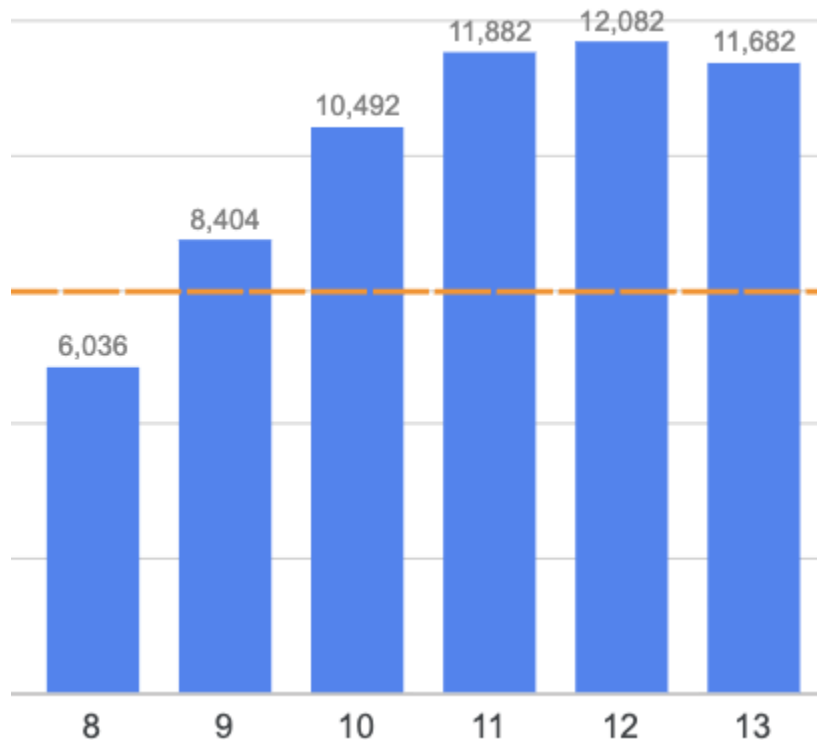
Hour	Orders	HoH (vs previous hour)
9:00	8,404	+39.2% vs 8:00
10:00	10,492	+24.9%
11:00	11,882	+13.2%

- Revenue: \$5,883,928 (17.1% of total)
- Orders increase continuously across 3 hours, building toward the lunch peak

Two interleaving behaviors: (a) Commuting to work (9:00) — browsing phones on public transit, reviewing cart, finalizing "incubated" orders. 9:00 surges +39.2% vs 8:00, aligning with US peak commute hours. (b) Work hours tab shopping (10:00–11:00) — "tab shopping" between work tasks with stable internet and larger screens.

This is the "upward ramp" leading to the lunch peak: 8,404 → 11,882 orders, increasing non-stop.

3.4. LUNCH BREAK — 12:00–13:00 (13.3%, 23,764 orders, AOV \$188.19) — FIRST PEAK



Hour	Orders	Revenue
12:00	12,082	\$2,316,821
13:00	11,682	\$2,155,390

- Total revenue: \$4,472,211 (13.0% of total)
- 12:00 is THE DAY'S SECOND HIGHEST hour (only behind 19:00)
- AOV \$188.19 — LOWEST among all 7 segments

The lowest AOV here is NOT because customers buy cheaper products — Laptops (34.1%) and Phones (26.0%) share is virtually identical to other time slots. The AOV fluctuation falls within the narrow \$188–\$198 range and reflects random variation per order, not a change in shopping behavior.

Behavior: Lunch break is the first official free time of the day. All browse activity from the morning (Wake & Browse + Commute & Work) accumulates and gets "checked out" here. Just 2 hours produce 23,764 orders — density of 11,882 orders/hour, 1.6x the average (7,434).

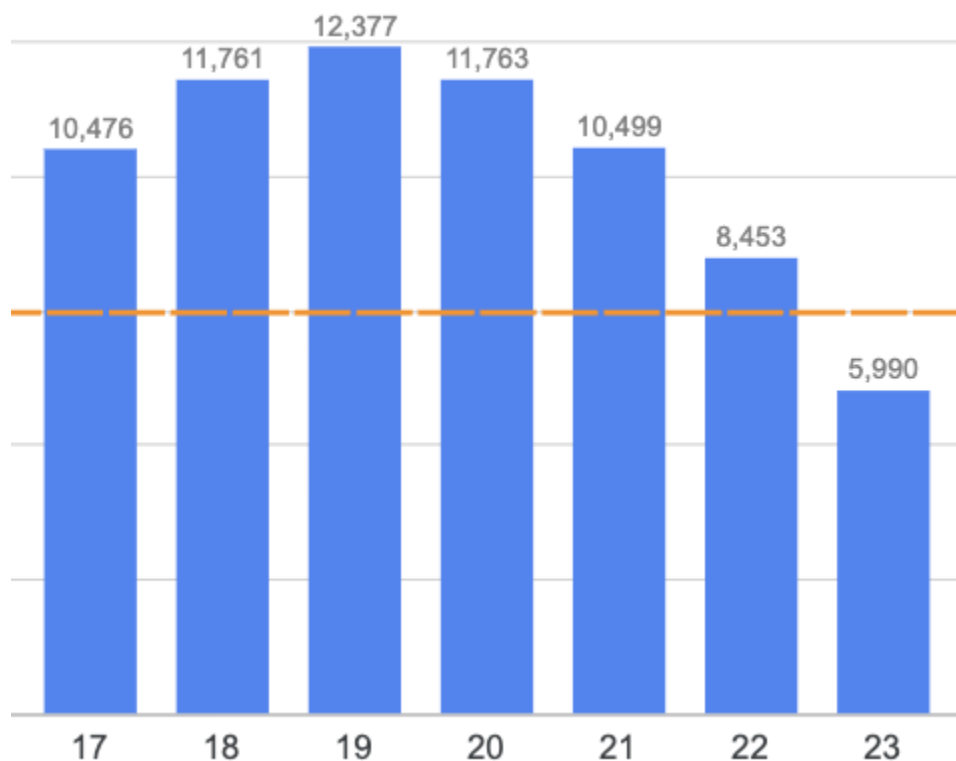
3.5. AFTERNOON PLATEAU — 14:00–17:00 (22.8%, 40,719 orders, AOV \$197.92)

- Decreases from the 13:00 peak but NEVER falls below average (7,434)
- Lowest point: 15:00 — 9,761 orders (still 31.3% above average)
- AOV \$197.92 — HIGHEST among all 7 segments
- Revenue: \$8,059,185 (23.4% of total)

The highest AOV of the day appears here, not in the evening. However, the gap is only \$4.63 vs Evening (\$193.02) — too small to conclude any meaningful difference. Product mix remains identical: Laptops 36.4%, Phones 25.2%.

Behavior: Customers return to work after lunch, less free time but can still "tab shop" between tasks. Orders decline but never collapse — this is the "bridge" between the 2 peaks, keeping e-commerce's "effective window" running continuously for 14 hours (9:00–22:00 all above average).

3.6. EVENING AT HOME — 18:00–21:00 (26.0%, 46,400 orders, AOV \$193.02) — HIGHEST PEAK



Hour	Orders	Revenue
18:00	11,761	\$2,219,348

19:00	12,377	\$2,412,939
20:00	11,763	\$2,281,716
21:00	10,499	\$2,042,001

- Total revenue: \$8,956,004 (26.0% of total) — highest among all 7 segments
- 19:00 = THE DAY'S ABSOLUTE PEAK (12,377 orders)
- AOV \$193.02 — near average, product mix stable (Laptops 35.0%, Phones 26.0%)

Why is this the highest peak? After 18:00, most have finished work. Home on the couch/computer, with the LONGEST free time of the day to browse, compare, and decide — no time pressure like lunch breaks. 4 consecutive hours (18–21h) all above 10,000 orders, creating the largest revenue block.

"Golden Hour" 19:00: The absolute peak — customers have finished dinner, begin their most focused online shopping session. 19:00 is the hottest hour in 5/7 days on the Heatmap.

3.7. LATE NIGHT WIND-DOWN — 22:00–1:00 (11.5%, 20,437 orders, AOV \$193.59)

- Rapid decline: 22:00 (8,453) → 0:00 (3,748) → 1:00 (2,246)
- But 22:00 is still above average (8,453 > 7,435)
- AOV \$193.59 — near average, product mix unchanged

Customers begin going to bed. "Bed-scrolling" behavior generates orders but gradually diminishes. Many browse but don't buy immediately — they save and purchase the next morning (linking to Wake & Browse at 6:00–8:00).

4. NOON PEAK vs EVENING PEAK — WHAT DOES THE DATA SHOW?

Comparing the 2 peak hours:

Metric	Noon Peak 12:00	Evening Peak 19:00	Difference
Orders	12,082	12,377	+295 orders
Revenue	\$2,316,821	\$2,412,939	+\$96,118
AOV	\$191.76	\$194.95	+\$3,19

Comparing the 2 expanded peak zones:

Metric	Noon Zone (10–13h)	Evening Zone (18–21h)	Difference
Total orders	46,138	46,400	+262 orders
Total revenue	\$8,717,108	\$8,956,004	+\$238,896
AOV	\$188.94	\$193.02	\$4.08

Product mix at the 2 peak zones:

SubCategory	Noon Zone	Evening Zone	Difference
Laptops	34.2%	35.0%	+0.8% ▾
Phones	26.4%	26.0%	-0.4% ▾
Monitors	18.6%	18.6%	0.0% ▾
Headphones	11.8%	11.4%	-0.4% ▾
TVs	4.2%	4.2%	0.0% ▾
Laundry	2.2%	2.4%	+0.2% ▾
Cables	1.9%	1.8%	-0.1% ▾
Batteries	0.6%	0.6%	0.0% ▾

RESULT: Product mix at the 2 peak zones is nearly IDENTICAL (max difference 0.8%). Customers buy the SAME ratio of Laptops, Phones, and accessories regardless of noon or evening.

The AOV gap (\$188.94 vs \$193.02 = \$4.08) is too small to conclude evening shoppers "buy more expensive." With identical product mix, the +\$238,896 revenue difference between the 2 zones comes purely from the evening zone having 262 MORE orders, not from higher order values.

The real difference between the 2 peaks lies in HOW THEY FORM:

- Noon zone: accumulated from 3 sequential behaviors (commute 9h → work 10-11h → lunch break 12-13h) — orders build gradually then burst at lunch

- Evening zone: burst from a single behavior — home, relaxed, with long uninterrupted free time (4 hours)

The 2 peak zones produce nearly equal order volume (0.6% difference), same product mix, same AOV range — but entirely different formation mechanisms.

5. HEATMAP ANALYSIS — DAY OF WEEK × HOUR

Heatmap Time × Day of the week

DayOfWeek	0	1	2	3	4	5	6	7	8	9	10	Hour / Order ID
Monday	543	304	164	121	119	184	337	561	868	1,193	1,534	
Tuesday	537	334	172	122	117	182	364	585	935	1,252	1,549	
Wednesday	531	327	197	124	103	208	332	547	814	1,173	1,465	
Thursday	514	321	160	122	124	171	338	539	850	1,256	1,519	
Friday	572	319	168	96	102	169	328	521	874	1,143	1,468	
Saturday	520	309	174	113	118	159	358	524	880	1,220	1,484	
Sunday	531	332	157	103	127	185	318	559	815	1,167	1,473	

Overall average: 1,062 orders per cell (per day-hour combination)

5.1. Most important finding: ABSOLUTE CONSISTENCY

Every day of the week has exactly 14 out of 24 hours above average. Monday = Tuesday = ... = Sunday. No exceptions.

WHY SO CONSISTENT? This is characteristic of e-commerce: unaffected by store hours, holidays, or street traffic. Online customers shop based on their PERSONAL LIFE RHYTHM (sleep/wake, work/rest), not store schedules. And this rhythm is virtually identical every day.

5.2. Top 5 hottest cells — 19:00 dominates:

1. Friday 19:00 — 1,810 orders (1.70x average)
2. Monday 19:00 — 1,796 orders (1.69x average)
3. Tuesday 18:00 — 1,788 orders (1.68x average)
4. Monday 12:00 — 1,770 orders (1.67x average)
5. Saturday 19:00 — 1,770 orders (1.67x average)

19:00 appears in 3 of the top 5 hottest cells, and is the peak hour in 5 out of 7 days. This is the "E-commerce Golden Hour" — the moment when customers are most psychologically ready to buy: done with work, done with dinner, relaxing.

5.3. Top 5 coldest cells — 3:00–4:00 AM always at the bottom:

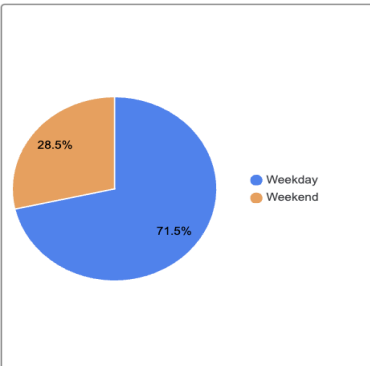
1. Friday 3:00 — 96 orders (0.09x average)
2. Sunday 3:00 — 103 orders
3. Wednesday 4:00 — 107 orders

4. Saturday 3:00 — 113 orders
5. Tuesday 4:00 — 117 orders

Hot/Cold ratio: $1,810 / 96 = 18.9x$ within the same day (Friday). E-commerce has an extremely large intraday fluctuation range.

5.4. Weekday vs Weekend — No Difference

Comparing Revenue Weekday vs We...



Metric	Weekday	Weekend
Avg orders/day	487	489
Avg revenue/day	\$94,105	\$94,582
Difference	—	+0.5%

WHY NO DIFFERENCE? With brick-and-mortar stores, weekends are typically busier because customers have free time to go shopping. But e-commerce breaks this pattern: online customers can buy at any time — a weekday lunch break is just as effective as a Sunday afternoon.

The 71.5% (Weekday) vs 28.5% (Weekend) split on the pie chart only reflects the 5:2 day ratio, NOT a behavioral difference.

6. CONNECTING HOURLY PATTERNS WITH PEAK MONTHS — DOES PRODUCT MIX CHANGE?

From the Overview analysis, we know monthly revenue fluctuates significantly (Jan \$1.82M → Dec \$4.61M). Two questions:

- Does the HOURLY PATTERN (Twin Peak) change by month?
- Does the PRODUCT MIX change by month?

6.1. Hourly pattern — does NOT change

The Twin Peak model (12h and 19h) remains consistent across all months. What changes during peak months (Oct–Dec) is the SCALE (more orders at every hour), not the SHAPE (still 2 peaks at the same time slots).

6.2. Product mix — ALSO does NOT change

SubCategory	January	April	October	December	Range
Laptops	33.9%	34.3%	36.2%	35.4%	2.3% ▾
Phones	27.9%	27.2%	25.2%	25.5%	2.7% ▾
Monitors	18.1%	18.4%	18.8%	18.8%	0.7% ▾
Headphones	11.1%	11.2%	11.4%	11.4%	0.3% ▾
TVs	4.0%	4.1%	4.2%	4.3%	0.3% ▾
Laundry	2.7%	2.5%	1.8%	2.2%	0.9% ▾
Cables	1.8%	1.8%	1.9%	1.8%	0.1% ▾
Batteries	0.6%	0.6%	0.6%	0.6%	0.0% ▾

From the lowest month (Jan) to the highest month (Dec), product mix barely changes. Laptops always ~34–36%, Phones ~25–28%, Top 3 (Laptops + Phones + Monitors) consistently ~80% of revenue.

6.3. Dual constant

Combining findings from Section 3 (product mix unchanged by HOUR) and Section 6 (unchanged by MONTH):

→ Product mix is a DUAL CONSTANT: unchanged across time of day, and unchanged across months of the year.

The only thing that changes is SCALE — peak months have more orders per hour, peak hours have more orders than off-peak hours — but the Laptops/Phones/Monitors/accessories ratio stays the same. This is the Volume-Driven characteristic, consistent with the Sales Overview conclusion.

7. KEY TAKEAWAYS & RECOMMENDATIONS

7.1. "TWIN PEAK" STRATEGY FOR E-COMMERCE: The two time slots 10–13h and 18–21h account for 51.9% of total orders in just 8 hours. All digital marketing campaigns (push notifications, emails, flash sales, social media ads) should be scheduled to concentrate on these 2 windows.

7.2. 19:00 IS THE "E-COMMERCE GOLDEN HOUR": The absolute peak of the day, hottest in 5 out of 7 days. All time-sensitive promotions (flash sales, limited deals, countdowns) should prioritize the 19:00 window. This is when customers are at home, relaxed, with the longest free time of the day.

7.3. PRODUCT MIX IS A CONSTANT — NO NEED TO DIFFERENTIATE PRODUCTS BY HOUR: Data shows SubCategory share is nearly IDENTICAL across all time slots (Laptops always ~35%, Phones ~26%). Customers buy the SAME ratio of products whether it's noon or evening. Product promotion strategy should focus on the HOURS WITH MOST ORDERS (18–21h) rather than trying to match specific products to specific hours.

7.4. NO SEPARATE WEEKDAY/WEEKEND STRATEGY NEEDED: Only a 0.5% difference between weekdays and weekends. E-commerce breaks the offline rule of "weekends are busier." The same marketing schedule works effectively every day.

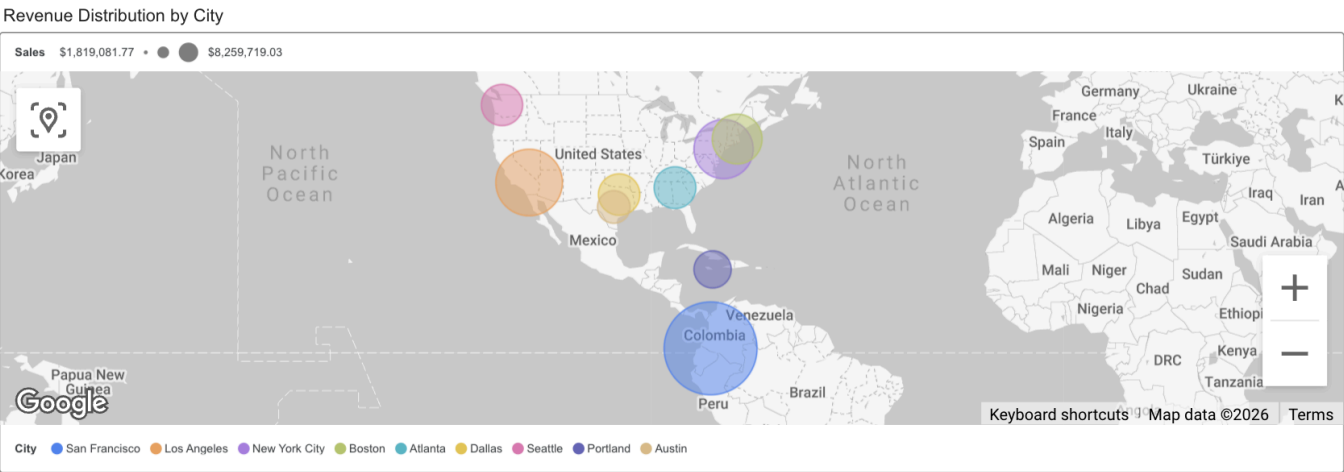
7.5. LEVERAGE "WAKE & BROWSE" 6–9h: Orders increase 254% in 3 hours — many are decisions "incubated" from the previous evening. Send push notifications/emails reminding customers of their "saved cart" at 7:00–8:00 AM to convert last night's browsers into this morning's buyers.

7.6. LOW-TRAFFIC HOURS 2–5h — LOW COST, HIGH EFFICIENCY: Still 4,061 orders/year during the "dead" window — an e-commerce exclusive. Run retargeting ads and scheduled emails during this window: advertising costs are cheapest (less competition), targeting the "night owl" group and customers in different time zones.

7.7. HOURLY PATTERN + PRODUCT MIX = DUAL CONSTANT: The Twin Peak shape doesn't change by month or season. Product mix doesn't change by hour or month. Time-based marketing strategy can be applied consistently year-round — only adjust BUDGET SCALE (increase in Q4), not the SCHEDULE or PRODUCT CATEGORIES promoted.

GEOGRAPHIC ANALYSIS — IN-DEPTH INSIGHT ANALYSIS

I. GEOGRAPHIC DISTRIBUTION OVERVIEW



Revenue is distributed across 9 cities in 3 major US geographic regions:

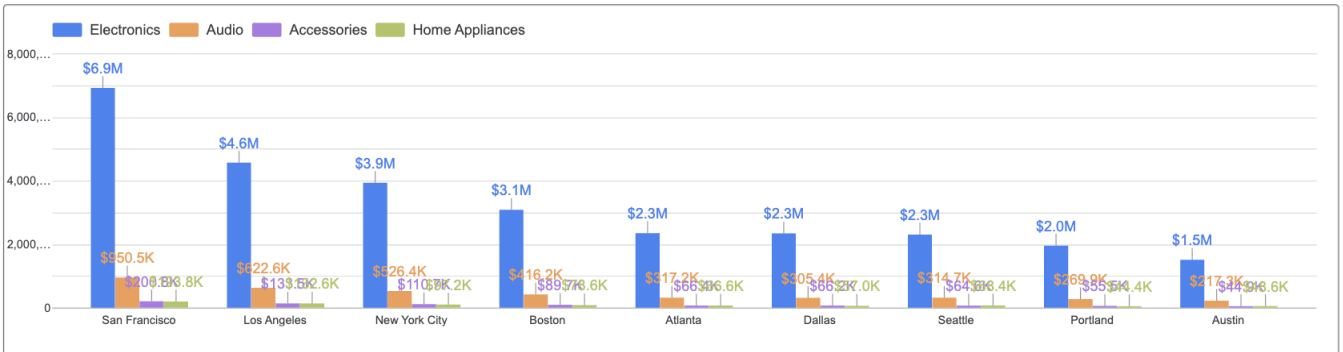
Region	Cities	Revenue	Share	Orders	AOV
West Coast	SF, LA, Seattle, Portland	\$18,783,021	54.5%	97,480	\$192.69
East Coast	NYC, Boston, Atlanta	\$11,121,458	32.2%	57,181	\$194.50
South/Central	Dallas, Austin	\$4,587,557	13.3%	23,745	\$193.20

Immediate finding from the overview table: AOV across the 3 regions is nearly identical (\$192.69 – \$194.50, spread of only \$1.81). The West Coast accounts for 54.5% of revenue — but also accounts for exactly 54.7% of total orders. The revenue gap between regions is entirely driven by ORDER VOLUME, not by customers in any region spending more per order.

7 out of 9 cities in the dataset are located on or near the coastline, contributing 86.7% of total revenue. Only 2 inland cities (Dallas, Austin) account for just 13.3%. This reflects US demographic characteristics: the largest metropolitan areas are concentrated along both coastlines — revenue distribution follows population density.

II. CITY RANKINGS — ALL 9 CITIES

Revenue Breakdown by City & Category



Rank	City	Revenue	Revenue %	Orders	Orders %	AOV
1	San Francisco	\$8,259,719	24.0%	42,887	24.0%	\$192.59
2	Los Angeles	\$5,451,571	15.8%	28,497	16.0%	\$191.30
3	New York City	\$4,662,977	13.5%	23,840	13.4%	\$195.59
4	Boston	\$3,660,315	10.6%	19,088	10.7%	\$191.76
5	Atlanta	\$2,795,499	8.1%	14,253	8.0%	\$196.13
6	Dallas	\$2,766,260	8.0%	14,238	8.0%	\$194.29
7	Seattle	\$2,747,605	8.0%	14,118	7.9%	\$194.62
8	Portland	\$2,320,338	6.7%	11,978	6.7%	\$193.72
9	Austin	\$1,819,082	5.3%	9,507	5.3%	\$191.34

Key findings from this table:

(a) Revenue % = Orders % is nearly perfect for every city. SF accounts for 24.0% of revenue and also exactly 24.0% of orders. Austin accounts for 5.3% of revenue and also exactly 5.3% of orders. This pattern holds across all 9 cities.

→ Larger cities do not spend more per order — they simply place more orders. This confirms the Volume-Driven model at the city level.

(b) AOV fluctuates within an extremely narrow range: \$191.30 (LA) → \$196.13 (Atlanta), a spread of only \$4.83. The city with the highest AOV (Atlanta) is actually the lowest-revenue city in the Top 5 — further evidence that AOV does not determine revenue.

(c) SF is 4.5x Austin (\$8.26M vs \$1.82M), yet AOV differs by just \$1.25 (\$192.59 vs \$191.34). The entire 4.5x gap comes from order count (42,887 vs 9,507).

III. SAN FRANCISCO — THE SOLE OUTLIER

SF accounts for 24.0% of national revenue — nearly 1/4 of total revenue from a single city. This is an unusually high concentration that warrants closer examination:

(a) SF vs the rest of the West Coast:

West Coast with SF: \$18,783,021 (54.5%) → West Coast dominates

West Coast WITHOUT SF: \$10,523,302 → LOSES to East Coast (\$11,121,458)

→ "West Coast leads" is actually "SF leads." Remove SF, and the two coasts are nearly equal.

(b) SF vs city groups:

SF (\$8.26M) > Bottom 4 combined (Dallas + Seattle + Portland + Austin = \$9.65M)

SF (\$8.26M) = 1.8x NYC (\$4.66M) — the most populous city in America

(c) Why is SF an outlier — what does the data show?

SF product mix: Laptops 35.0%, Phones 26.3%, Monitors 18.2% — IDENTICAL to other cities (spread < 2%)

SF AOV: \$192.59 — mid-table, not higher

SF top 5 products: AAA Batteries, AA Batteries, USB-C Cable, Lightning Cable, Wired Headphones — IDENTICAL to all 9 cities

→ SF doesn't buy DIFFERENTLY — SF buys MORE. SF customers purchase the same products, at the same prices, in the same category proportions as every other city — only the order volume is 4.5x higher than the smallest city.

SF Bay Area is the center of Silicon Valley — where the density of technology users is unusually high relative to population size. The data confirms: high density creates high volume, not different purchasing behavior.

IV. NATIONWIDE UNIFORMITY — THE CORE FINDING

Category / SubCategory Sales Heatmap by City

City	Accessories				Electronics		Audio	Home Appliances
	Batteries	Cables	Monitors	Phones	Laptops	TVs	Headphones	Laundry
San Francisco	13,962	11,450	5,716	3,511	2,099	1,154	12,011	323
Los Angeles	9,405	7,554	3,839	2,283	1,391	727	7,868	221
New York City	7,751	6,304	3,342	1,918	1,217	628	6,594	162
Boston	6,477	5,050	2,537	1,560	925	553	5,291	131
Dallas	4,765	3,715	1,923	1,158	726	420	3,926	95
Atlanta	4,552	3,794	1,905	1,168	736	407	3,929	111
Seattle	4,625	3,644	1,974	1,159	689	375	3,972	114
Portland	4,018	3,113	1,645	929	611	312	3,349	74
Austin	3,092	2,558	1,234	759	461	242	2,724	81

This is the most important finding of the geographic analysis: purchasing behavior is COMPLETELY UNIFORM nationwide. Evidence from 4 dimensions:

(a) SubCategory Revenue Share — uniform across all cities:

SubCategory	Min	Max	Spread
Laptops	34.1% (Seattle)	36.5% (Portland)	2.4%
Phones	24.9% (Portland)	26.5% (Boston)	1.6%
Monitors	17.9% (Atlanta)	19.5% (Seattle)	1.6%
Headphones	11.0% (Dallas)	11.9% (Austin)	0.9%
TVs	4.0% (multiple)	4.6% (Dallas)	0.6%
Laundry	1.9% (Portland)	2.7% (Austin)	0.8%
Cables	1.8% (multiple)	1.9% (multiple)	0.1%
Batteries	0.6% (all)	0.6% (all)	0.0%

Maximum spread is only 2.4% (Laptops). No city has a distinctive product structure.

(b) Quantity Share — same pattern:

SubCategory	Min	Max	Spread
Batteries	27.4% (Atlanta)	28.8% (Boston)	1.4%
Headphones	23.5%	24.4% (Austin)	0.9%

	(Boston/Dallas)		
Cables	22.0% (Seattle)	22.9% (Atlanta)	0.9%
Monitors	11.1% (Austin)	12.0% (NYC)	0.9%

Both by revenue and by quantity, SubCategory shares are nearly identical across all 9 cities.

(c) Top 5 best-selling products — IDENTICAL across all 9 cities:

All 9 cities share the same top 5 by quantity sold:

- AAA Batteries (4-pack)
- AA Batteries (4-pack)
- USB-C Charging Cable
- Lightning Charging Cable
- Wired Headphones

No exceptions — from SF (largest city) to Austin (smallest), customers buy the same products in the same priority order.

(d) AOV by SubCategory — uniform:

SubCategory	Min	Max	Spread
Laptops	\$1,362 (ATL)	\$1,386 (POR)	\$24 (1.7%)
Phones	\$617 (SEA)	\$625 (POR)	\$8 (1.3%)
Monitors	\$264 (LA)	\$273 (SEA)	\$9 (3.4%)
Headphones	\$82 (DAL)	\$84 (ATL/POR)	\$2 (2.4%)
Laundry	\$600	\$600	\$0 (0.0%)

Customers in every city pay the same average price for the same product types.

→ Conclusion: Product demand is NATIONWIDE — unaffected by local factors. Whether coastal or inland, West Coast or East Coast, customers buy the same products, in the same proportions, at the same prices. The only difference between cities is the NUMBER of orders.

V. FULLY SYNCHRONIZED SEASONALITY

(a) All 9 cities have their lowest month in January and highest in December — no exceptions.

(b) Q4 (Oct–Dec) accounts for nearly identical shares across all cities:

City	Q4 Share
Seattle	34.8%
Dallas	34.4%
Atlanta	33.7%
Boston	33.6%
New York City	33.5%
Austin	33.4%
San Francisco	33.2%
Portland	33.0%
Los Angeles	32.9%

Spread of only 1.9% (32.9% – 34.8%). 1/4 of the year (3/12 months) generates 1/3 of revenue — and this ratio is uniform across all cities.

(c) Highest month / lowest month ratio (Dec/Jan):

City	Dec/Jan Ratio
Seattle	2.76x
Dallas	2.69x
Austin	2.67x
Portland	2.64x
San Francisco	2.56x
Boston	2.55x
New York City	2.49x
Atlanta	2.42x
Los Angeles	2.38x

All cities fluctuate within a 2.38x – 2.76x range. No city has a unique peak season or an out-of-phase cycle.

The year-end holiday season (Black Friday, Christmas, New Year) is a nationwide shopping event in the US — the data confirms its impact is uniform across all 9 cities, regardless of region.

→ The marketing calendar can be applied NATIONWIDE — Q4 push for all cities simultaneously, no need for region-specific seasonal campaigns.

VI. CONCENTRATION & RISK

Revenue is highly concentrated following a Pareto pattern:

Group	Revenue	Share
Top 1 (SF)	\$8,259,719	24.0%
Top 3 (SF + LA + NYC)	\$18,374,267	53.3%
Top 5 (+ Boston + Atlanta)	\$24,830,081	72.0%
Bottom 4 (DAL+SEA+POR+AUS)	\$9,653,285	28.0%

More than half of revenue comes from 3 cities. SF alone accounts for nearly 1/4.

Concentration risk:

SF declines 20% → loss of ~\$1,652,000 ≈ Austin's entire annual revenue

SF + LA decline 15% → loss of ~\$2,057,000 ≈ Portland + Austin combined revenue

Top 3 at 53.3% means that a downturn in just 3/9 cities would impact more than half of total revenue

However, the Bottom 4 still account for 28.0% (\$9.65M) — these smaller markets should not be overlooked.

Growth potential within the Bottom 4 is not evenly distributed. Since purchasing behavior has been proven uniform nationwide — same products, same AOV, same conversion patterns — metro population is the best proxy for estimating the true addressable market still left untapped:

City	Metro Population	Revenue	Assessment
Dallas	~7.8M	\$2.77M	Most underserved
Seattle	~4.0M	\$2.75M	Medium potential

Portland	~2.5M	\$2.32M	Proportional to size
Austin	~2.3M	\$1.82M	Proportional to size

Dallas stands out the most: metro population 3x larger than Portland and Austin, yet revenue sits in the same bottom group. This is not a signal of a weak market — it is a signal of a market that has not been sufficiently reached.

The current playbook has been proven to work in SF, LA, and NYC. No changes to products or strategy are needed — simply scaling reach in Dallas can unlock revenue that is currently being left on the table.

→ Within the Bottom 4: prioritize Dallas first, then Seattle.

VII. CONCLUSIONS & RECOMMENDATIONS

7.1. ALLOCATE BUDGET BY REVENUE SHARE

Revenue share is the definitive metric for investment priority. Since purchasing behavior is uniform across all cities, marketing effectiveness depends on market SCALE:

A 10% increase in SF (\$826K) has greater impact than a 45% increase in Austin (\$819K)
 SF + LA (California) = 39.8% of revenue → a California-focused campaign covers nearly 2/5 of total revenue

NYC = 42% of East Coast revenue → the most efficient gateway to grow the Eastern region

7.2. ONE NATIONWIDE PRODUCT STRATEGY

Data proves: same product mix, same AOV, same top 5 products across all 9 cities. No need for region-specific catalogs or localized product strategies — one playbook applies nationwide, with only budget allocation adjusted by market size.

7.3. ONE NATIONWIDE MARKETING CALENDAR

Seasonality is 100% synchronized: all cities share the same January low, December high, and ~33% Q4 share. Q4 campaigns deploy simultaneously across all markets — no need for region-specific seasonal schedules.

7.4. REDUCE CONCENTRATION RISK — PRIORITIZE DALLAS

SF at 24% is overly dominant. Long-term strategy must drive growth in tier 2 cities to reduce dependence on a single market.

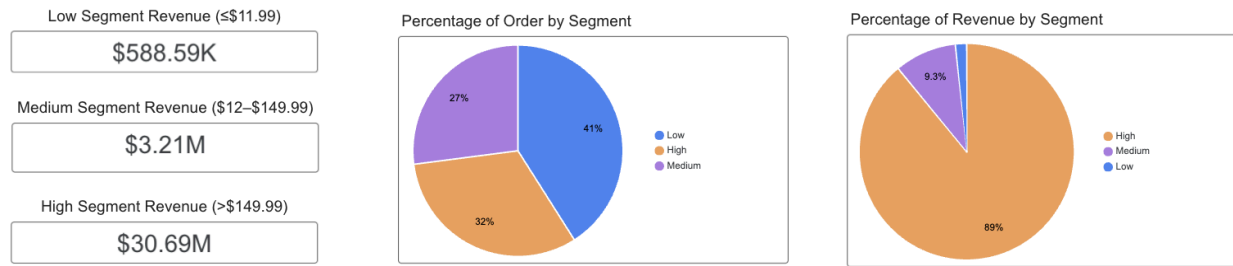
Within the Bottom 4, Dallas is the #1 priority: metro population of ~7.8M (3x Portland/Austin) yet revenue sits level with the group — a clear signal of a market that has not been sufficiently reached. The playbook proven at SF/LA/NYC requires no changes — simply scale reach in Dallas to unlock revenue currently being left on the table. After Dallas, prioritize Seattle (~4.0M metro population, medium potential).

VIII. SUMMARY: WHAT DOES GEOGRAPHY ACTUALLY AFFECT?

Question	Answer	Evidence
Does geography affect REVENUE?	Yes — Scale	SF = 24%, Austin = 5.3% (4.5x gap)
Does geography affect WHICH PRODUCTS customers buy?	No	SubCategory share spread max 2.4%
Does geography affect SPENDING LEVEL?	No	AOV \$191–\$196, spread \$4.83 across 9 cities
Does geography affect PRODUCT PRICING?	No	AOV per SubCategory spread 0–3.4%
Does geography affect SEASONALITY?	No	All peak in December, Q4 share 32.9–34.8%

→ Geography determines only SCALE (how many orders), not BEHAVIOR (what customers buy, how much they spend, when they buy). The business model is completely uniform geographically — all prior findings (Volume-Driven Business, Product Mix Constant, 3-Phase Revenue Cycle) hold true in every city.

ORDER SEGMENTATION & PRICING — INSIGHT ANALYSIS



I. ORDER SEGMENT ANALYSIS

Orders are divided into 3 segments based on order value:

Segment	Value Threshold	Orders	Order Share	Revenue	Revenue Share
Low	$\leq \$11.99$	73,103	41.0%	\$588,658	1.7%
Medium	\$11.99 – \$149.99	48,263	27.0%	\$3,212,885	9.3%
High	$> \$149.99$	57,071	32.0%	\$30,690,493	89.0%

Critical Finding — "The 41% vs 1.7% Paradox":

- The Low segment accounts for the LARGEST SHARE OF ORDERS (41%) but contributes only 1.7% of revenue
- The High segment accounts for 32% of orders but generates 89% of revenue
- In other words: for every 10 orders, 4 belong to the Low segment, but these 10 orders bring in only \$1.7 out of every \$100 in revenue
- This is a classic "top-heavy" model: revenue is extremely dependent on high-priced products

II. EASIEST TO SELL — WHICH PRICE POINTS SELL BEST?

Ranked by Quantity Sold:

Rank	Product	Price	Qty Sold	Characteristic
1	AAA Batteries (4-pack)	\$2.99	31,017	Cheapest, best-selling

2	AA Batteries (4-pack)	\$3.84	27,635	Essential item
3	USB-C Charging Cable	\$11.95	23,975	Universal accessory
4	Lightning Charging Cable	\$14.95	23,217	Apple accessory
5	Wired Headphones	\$11.99	20,557	Budget headphones

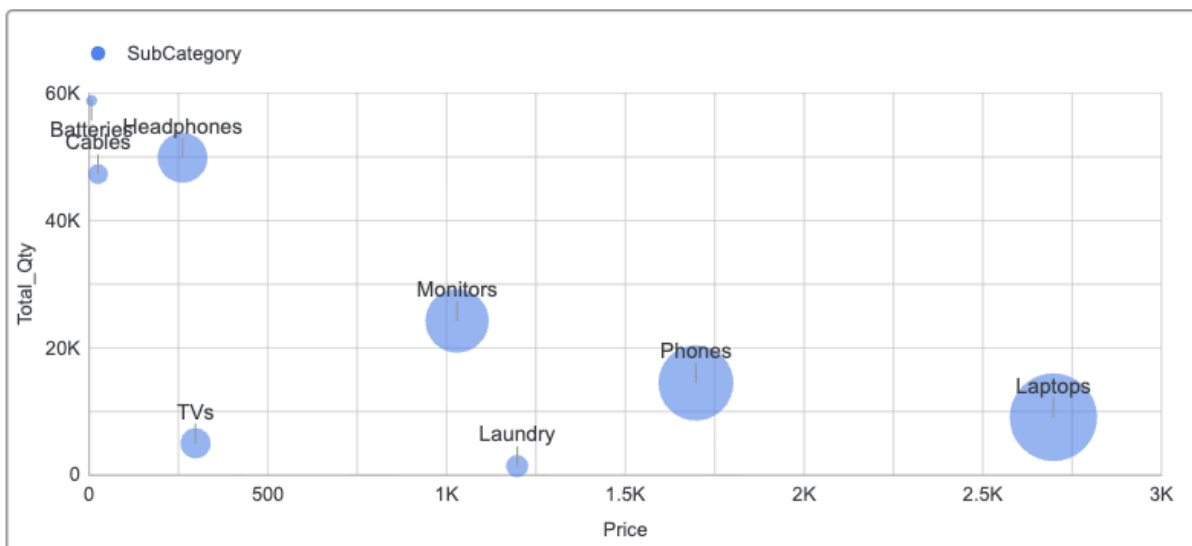
Key Findings:

- All Top 5 best-selling products are priced UNDER \$15 — low-cost, consumable accessories
- Common trait: fast-moving consumer goods (batteries die, cables break, headphones degrade), customers purchase repeatedly
- These 5 products sold 126,401 units but generated only \$1,078,933 in revenue (3.1%) — very high volume but low value
- Purchase barrier is nearly zero: low price → no deliberation needed → fast buying decisions

Price-Quantity Rule:

- Correlation coefficient (Price vs Quantity) = -0.60 (strong negative correlation)
- As price increases → quantity sold decreases significantly
- However, some products "break the rule" (see Section IV)

Price vs Quantity Sold by SubCategory



III. HARDEST TO SELL — WHICH PRICE POINTS CREATE BARRIERS?

Ranked by LOWEST Quantity Sold:

Rank	Product	Price	Qty Sold	Core Issue
19	LG Dryer	\$600	646	High price + niche demand
18	LG Washing Machine	\$600	666	High price + niche demand
17	Vareebadd Phone	\$400	2,068	Weak brand
16	20in Monitor	\$109.99	4,129	Small size, outdated
15	ThinkPad Laptop	\$999.99	4,130	High price but sells steadily

Detailed Analysis by "Hard-to-Sell" Group:

3.1. LG Washing Machine & LG Dryer — "Not E-Commerce's Playground"

- Both priced at \$600, selling only ~650 units/year
- These are large home appliances — customers typically buy at physical stores for hands-on inspection
- Extremely low repeat purchase frequency (buy once, use for 5-10 years)
- Niche market: only purchased by people moving to new homes or replacing broken units

3.2. Vareebadd Phone — "Stuck in the Middle"

- Priced at \$400, sold 2,068 units — fewer than Google Phone (\$600, 5,532 units) despite being \$200 CHEAPER
- Paradox: lower price but fewer sales → the problem isn't price, it's BRAND
- Vareebadd is stuck in the middle: more expensive than accessories but lacks the brand power of iPhone/Google
- Customers are willing to pay an extra \$200-300 for iPhone or Google Phone rather than choosing Vareebadd

3.3. 20in Monitor — "An Outdated Product"

- Priced at \$109.99 but sold only 4,129 — far behind the 27in FHD (\$149.99, 7,550 units)
- Customers are willing to pay an extra \$40 for a screen 7 inches larger
- The 20-inch monitor is being phased out by the market as work-from-home trends demand larger screens

IV. "RULE-BREAKERS" — HIGH PRICE BUT STILL SELLING STRONG

While the general rule is "high price = fewer sales," 4 products stand out:

Product	Price	Qty Sold	Why It Breaks the Rule
Apple AirPods Headphones	\$150	15,661	Brand power + Apple ecosystem
iPhone	\$700	6,849	Flagship, inelastic demand
27in 4K Gaming Monitor	\$389.99	6,244	Strong gaming niche
34in Ultrawide Monitor	\$379.99	6,199	WFH trend, productivity

Analysis:

4.1. Apple AirPods (\$150) — "The Ecosystem Effect"

- Sold 15,661 units — MORE than any single Electronics product
- Priced 12.5x higher than Wired Headphones (\$11.99) but sold nearly the same volume (15,661 vs 20,557)
- Apple's brand strength + ecosystem integration (iPhone + AirPods) eliminates price as a barrier

4.2. iPhone (\$700) — "Inelastic Demand"

- Sold 6,849 units — more than Google Phone (\$600) despite being \$100 MORE EXPENSIVE
- Customers buy iPhone for the brand and ecosystem, not the price
- Proves: in the phone segment, brand loyalty > price sensitivity

4.3. Premium Monitor Pair (\$380-\$390) — "The WFH Trend"

- 27in 4K Gaming (6,244) and 34in Ultrawide (6,199) sell at nearly identical volumes
- Priced 3.5x higher than 20in Monitor (\$109.99) but sell MORE (6,200 vs 4,129)
- Customers prioritize screen quality for gaming and working from home

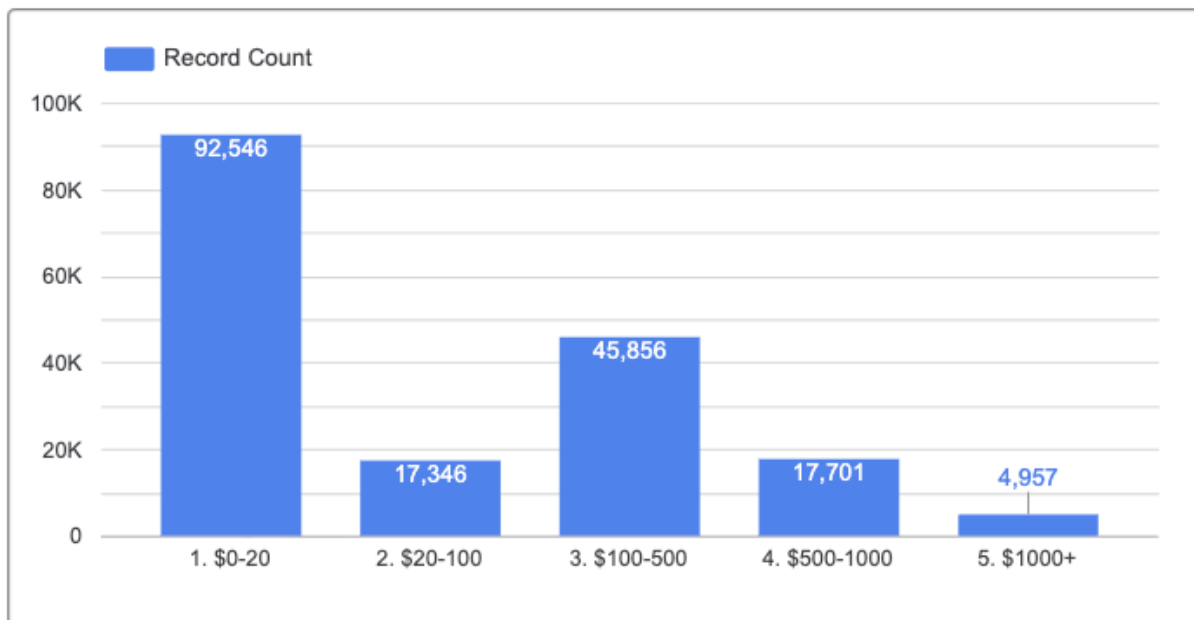
V. PRICE vs REVENUE ANALYSIS: THE GREAT PARADOX

Correlation coefficient (Price vs Revenue) = +0.87 (very strong positive correlation)

The Paradox: Higher price → fewer units sold, BUT higher price → MUCH MORE revenue

Price Band	Qty Sold	Revenue Share	Observation
\$0–20	126,401	3.1%	Best-selling, lowest revenue
\$20–100	13,457	3.9%	10x fewer units but same revenue
\$100–500	46,670	31.9%	Best balance zone
\$500–1000	17,823	37.8%	Primary revenue driver
\$1000+	4,728	23.3%	Fewest units but massive revenue

Order Distribution by Price Range



Discovery: The "Golden Price Zone" is \$100–\$500

- The \$100-500 range achieves the best balance between sales volume (46,670) and revenue (31.9%)
- This is the price range where customers are still willing to spend without excessive deliberation
- Includes: AirPods (\$150), Monitors (\$110-\$390), Flatscreen TV (\$300)

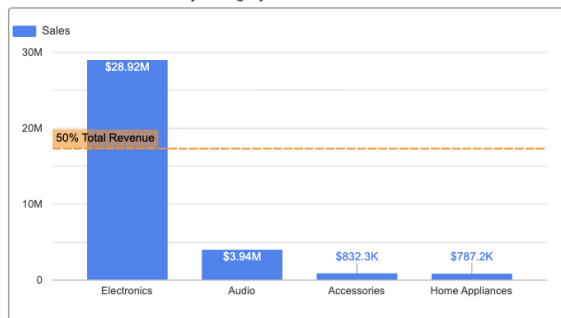
VI. REVENUE CONCENTRATION — THE PARETO RULE

6.1. Concentration by Category:

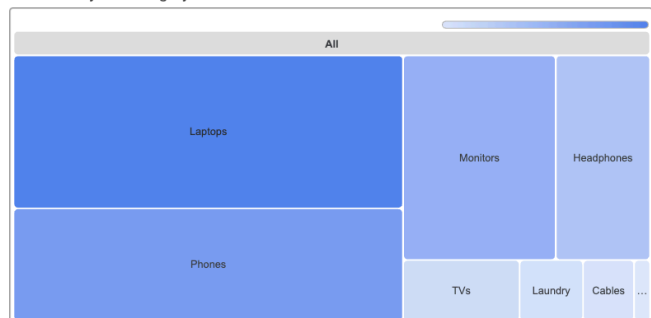
Category	Revenue	Share	Cumulative
Electronics	\$28,923,618	83.9%	83.9%
Audio	\$3,940,246	11.4%	95.3%
Accessories	\$832,302	2.4%	97.7%
Home Appliances	\$787,200	2.3%	100.0%

→ Electronics ALONE accounts for 83.9% of total revenue — nearly double the 50% threshold. The remaining three categories combined contribute only 16.1%. This is an extremely high concentration: the entire business is almost entirely dependent on ONE category.

Revenue Concentration by Category



Revenue by Subcategory



6.2. Concentration by Product (within Electronics):

- Top 3 products (Macbook, iPhone, ThinkPad) = 49.2% of revenue
- Top 7 products = 79.5% of revenue
- Top 10 products = 90.9% of revenue
- Remaining 9 products contribute only 9.1%

Pareto Variant: 37% of products (7/19) generate 80% of revenue.

6.3. Concentration Risk:

- Category level: Electronics declines 10% → loss of ~\$2.89M ≈ the entire Audio category revenue
- Product level: Macbook Pro (\$8.04M = 23.3%) declines 20% → loss of \$1.6M ≈ Portland's entire revenue
- Dual risk: revenue depends on 1 category (Electronics 83.9%), and within that category, depends on 1 product (Macbook 23.3%)

VII. PRICING OPTIMIZATION STRATEGY

7.1. Bundling Strategy — "Pull Low Up to Medium"

- 41% of orders belong to the Low segment ($\leq \$11.99$), buying only cheap accessories
- Opportunity: Bundle accessories with main products to increase order value
- Examples:
 - + iPhone (\$700) + Lightning Cable (\$14.95) + AirPods (\$150) = Bundle \$829 (save \$35.95)
 - + Google Phone (\$600) + USB-C Cable (\$11.95) = Bundle \$599 (save \$12.95)
- Target: Convert 10% of Low orders to Medium → increase revenue by ~\$260K

7.2. Upsell Strategy — "Push Medium Up to High"

- Customers buying Bose Headphones (\$99.99) → upsell to Apple AirPods (\$150)
 - + Only \$50 difference but 50% more revenue per order
- Customers buying 20in Monitor (\$109.99) → upsell to 27in FHD (\$149.99)
 - + \$40 difference for 7 inches larger — very compelling
- Customers buying Google Phone (\$600) → upsell to iPhone (\$700)
 - + Strong brand loyalty, \$100 difference is not a major barrier

7.3. Strategy for "Stuck-in-the-Middle" Products

- Vairebadd Phone (\$400): Reduce price to \$349 or bundle with free accessories to create perceived value
- 20in Monitor (\$109.99): Consider discontinuing or aggressively discounting to clear inventory

7.4. Strategy for Home Appliances

- LG Washing Machine & Dryer (\$600 each): Sell as a "Washer + Dryer" combo at \$1,099 (save \$101)
- Focus marketing on customers who are moving to new homes or renovating

7.5. Protect "Cash Cow" Products

- Macbook Pro (\$1,700) contributes 23.3% of revenue — DO NOT discount
- iPhone (\$700) contributes 13.9% — maintain pricing, focus on purchasing experience
- These two products are "cash cows" — protect margins, avoid price wars

VIII. KEY CONCLUSIONS

1. REVENUE DEPENDS ON HIGH-PRICED PRODUCTS: 89% of revenue comes from the High segment ($> \$149.99$), despite only 32% of orders. Each High segment order generates \$537 — 67 times a Low order (\$8). This is why 32% of orders generate 89% of revenue.

2. LOW-COST ACCESSORIES = TRAFFIC DRIVERS: 41% of orders purchase accessories under \$12 — these are not a revenue source but a customer acquisition channel. Use accessories as an "entry point" to cross-sell high-value products.

3. BRAND > PRICE: iPhone outsells Google Phone despite being \$100 more expensive. AirPods sell nearly as much as Wired Headphones despite being 12.5x the price. Brand power overcomes price barriers.

4. "GOLDEN PRICE ZONE" \$100-\$500: This is the optimal equilibrium between volume and revenue. Expanding the product catalog in this price range will maximize both metrics.

5. ELIMINATE WEAK PRODUCTS: 20in Monitor and Vareebadd Phone are underperforming — decisions needed on aggressive discounting, bundling, or discontinuation to reallocate resources.

6. CONCENTRATION RISK: Top 3 products = 49.2% of revenue. Revenue diversification is needed to reduce dependency on Macbook Pro.

ACTION PRIORITY ORDER: (1) Protect cash cows — Macbook + iPhone (Section 7.5), (2) Exploit the Golden Price Zone \$100-\$500 (Section 5), (3) Convert Low segment to Medium via bundles (Section 7.1), (4) Eliminate weak products (Section 7.3).

MARKET BASKET ANALYSIS — INSIGHT ANALYSIS

I. BASKET OVERVIEW

Total Orders	Multi-product Orders	Multi_Product_Rate	Single-product Orders	Single_Product_Rate
178,406	7,133	4.0%	171,273	96.00%

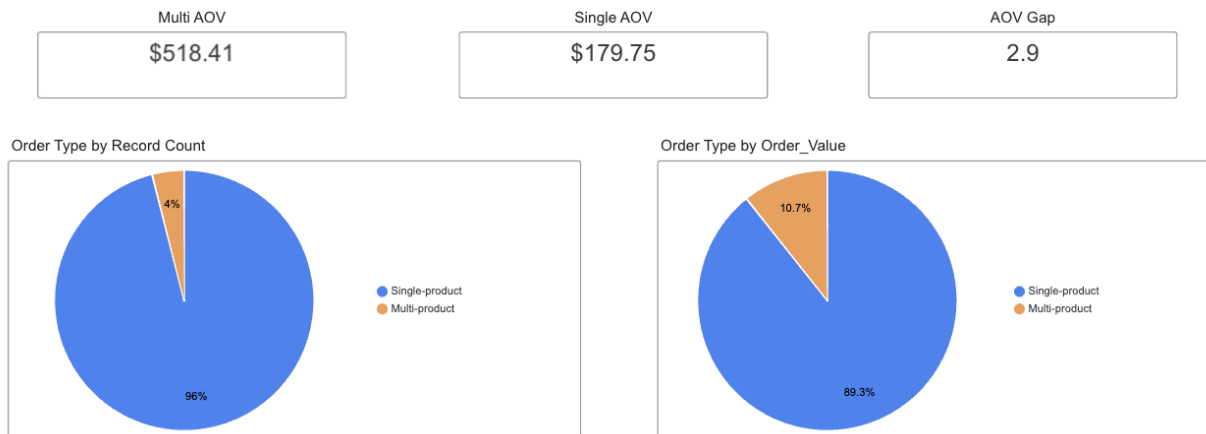
Metric	Value
Total Orders	178,406
Multi-product Orders	7,133
Multi-product Rate	4.0%
Single-product Orders	171,273

Opening Insight:

- 96% of orders contain only 1 product — an extremely high ratio
- Only 4% of orders (7,133) purchase ≥ 2 products simultaneously
- Cross-buying behavior is very rare, but the product pairs within this 4% carry highly valuable signals

II. REVENUE IMPACT OF CROSS-BUYING

2.1. Revenue: Multi-product vs Single-product Orders



	Multi-product	Single-product
Orders	7,133 (4.0%)	171,273 (96.0%)
Revenue	\$3,697,785 (10.7%)	\$30,785,581 (89.3%)
AOV	\$518.41	\$179.75

→ 4% of orders generate 10.7% of revenue. Multi-product AOV is 2.9x single-product (\$518 vs \$180). Each multi-product order delivers \$339 more value than single-product.

2.2. Add-on Value by Anchor Product

Anchor = highest-priced product in a multi-product order.

Anchor	Orders	Price	Order AOV	Add-on Value
iPhone	1,850	\$700.00	\$755.84	\$55.84
Google Phone	1,612	\$600.00	\$635.68	\$35.68
Vareebadd Phone	585	\$400.00	\$436.64	\$36.64
Macbook Pro	193	\$1,700.00	\$1,920.23	\$220.23
27in 4K Monitor	207	\$389.99	\$456.72	\$66.73

→ Macbook Pro triggers \$220 add-on per order — the highest add-on value. iPhone is the most popular anchor (1,850 orders) but add-on is only \$56. Two priority paths: iPhone for volume, Macbook for value.

2.3. Revenue Uplift Scenarios

If multi-product rate increases from the current 4%:

Target Rate	Orders to Convert	Add. Revenue	% Total Rev
5% (+1%)	1,787	+\$605,000	+1.8%
6% (+2%)	3,571	+\$1,209,000	+3.5%
8% (+4%)	7,139	+\$2,418,000	+7.0%
10% (+6%)	10,707	+\$3,626,000	+10.5%

→ Just a 2% increase (from 4% → 6%) generates \$1.21M in additional revenue — equivalent to 3.5% of current total revenue.

Note: Scenarios assume converted orders achieve AOV equivalent to the current multi-product average (\$518). Actual results may be lower — estimated real revenue uplift is in the range of 70–100% of the figures above.

III. PRODUCT PAIR ANALYSIS & ASSOCIATION METRICS

Among 50 product pairs analyzed, ONLY 3 PAIRS have Lift > 1 (green-highlighted on dashboard). Lift > 1 means buying A increases the likelihood of buying B above normal — a genuine relationship.

Association Rules Detail

	Product_A	Product_B	Co_Occurrence	Confidence_A_to_B	Confidence_B_to_A	Lift
1.	Google Phone	USB-C Charging Cable	997	18.06%	4.56%	1.47
2.	USB-C Charging Cable	Vareebadd Phone	368	1.68%	17.82%	1.46
3.	Lightning Charging Cable	iPhone	1,011	4.68%	14.78%	1.22
4.	Google Phone	Wired Headphones	422	7.64%	2.24%	0.72
5.	Vareebadd Phone	Wired Headphones	149	7.22%	0.79%	0.68
6.	Wired Headphones	iPhone	462	2.45%	6.75%	0.64
7.	Apple Airpods Headphones	iPhone	373	2.4%	5.45%	0.63
8.	Bose SoundSport Headphones	Google Phone	228	1.71%	4.13%	0.55
9.	Bose SoundSport Headphones	Vareebadd Phone	82	0.62%	3.97%	0.53
10.	USB-C Charging Cable	Wired Headphones	203	0.93%	1.08%	0.09

Anchor	Target	Co-occ	Conf	Lift	Mechanism
Google Phone	USB-C Cable	997	18.1%	1.47	Same USB-C port
Vareebadd Phone	USB-C Cable	368	17.8%	1.46	Same USB-C port
iPhone	Lightning Cable	1,011	14.8%	1.22	Lightning port

Rule: The Anchor is always a PHONE, the Target is always a CHARGING CABLE.
One-directional relationship: Phone → Cable.

All 47 remaining pairs have Lift < 1 — including surprises:

Product Pair	Co-occ	Lift	Observation
Wired Headphones + iPhone	462	0.64	High volume but low Lift
Google Phone + Headphones	422	0.72	Buy Phone rarely add headphones
Apple AirPods + iPhone	373	0.63	Same Apple but SUPPRESS
USB-C + Wired Headphones	203	0.09	Nearly independent

Key Finding — Apple AirPods + iPhone (Lift 0.63):

Airpods (\$150) sold 15,661 units — has become a standalone product, no longer dependent on iPhone.

CONCLUSION: Cross-buying behavior occurs almost exclusively when there is a DIRECT FUNCTIONAL NEED (needing a cable to charge the phone). No evidence of impulse buying or product exploration.

IV. ACTIONABILITY MATRIX

Classifying 50 product pairs into 3 priority tiers based on Lift, Co-occurrence, and practical value.

TIER 1 — ACT NOW (Lift > 1, proven functional relationship)

Product Pair	Co-occ	Lift	Conf	Action
Lightning Cable + iPhone	1,011	1.22	14.8%	Bundle on page
Google Phone + USB-C	997	1.47	18.1%	Bundle on page
USB-C + Vareebadd Phone	368	1.46	17.8%	Bundle on page

→ The only 3 pairs with Lift > 1. All Phone + Cable — deploy immediately with "Frequently bought together" or combo discount.

TIER 2 — TEST & LEARN (Co-occurrence > 200, Lift 0.5–1.0)

Product Pair	Co-occ	Lift	Conf	Action
Wired Headphones + iPhone	462	0.64	6.8%	Suggest at checkout
Google Phone + Headphones	422	0.72	7.6%	Suggest at checkout
Apple Airpods + iPhone	373	0.63	5.5%	Upsell headphones
Bose + Google Phone	228	0.55	4.1%	Upsell headphones

→ Sufficient volume but Lift < 1. Requires A/B testing at checkout, measure conversion over 2–4 weeks.

TIER 3 — MONITOR (Co-occurrence < 200 or Lift < 0.1)

Group	Pairs	Avg Lift	Observation
Accessory + Accessory	25	0.04	Cable + Battery, Cable + Headphone
Monitor + Accessory	15	0.04	Volume too low (25–43)

Accessory + Premium	3	0.04	Battery/Cable + iPhone/ThinkPad
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→ 43 remaining pairs lack sufficient signal. No resource investment, monitor only.

Matrix Summary: Tier 1: 3 pairs (6%) — deploy now. Tier 2: 4 pairs (8%) — A/B test. Tier 3: 43 pairs (86%) — no action.

V. MISSED OPPORTUNITIES

5.1. Multi-product Rate by Product

Product	Orders	Multi	Multi Rate	Price	Revenue
Google Phone	5,521	1,633	29.6%	\$600	\$3,319,200
Vareebadd Phone	2,064	601	29.1%	\$400	\$827,200
iPhone	6,838	1,862	27.2%	\$700	\$4,794,300
USB-C Cable	21,851	2,017	9.2%	\$11.95	\$286,501
Wired Headphones	18,846	1,605	8.5%	\$11.99	\$246,478
Lightning Cable	21,599	1,720	8.0%	\$14.95	\$347,094
Apple Airpods	15,521	925	6.0%	\$150	\$2,349,150
Bose SoundSport	13,295	765	5.8%	\$99.99	\$1,345,565
LG Washing Machine	666	28	4.2%	\$600	\$399,600
ThinkPad Laptop	4,124	172	4.2%	\$999.99	\$4,129,959
Macbook Pro	4,720	190	4.0%	\$1,700	\$8,037,600
27in 4K Monitor	6,224	241	3.9%	\$389.99	\$2,435,098
LG Dryer	646	25	3.9%	\$600	\$387,600
34in Ultrawide	6,174	232	3.8%	\$379.99	\$2,355,558
27in FHD	7,495	276	3.7%	\$149.99	\$1,132,424

Monitor					
20in Monitor	4,096	153	3.7%	\$109.99	\$454,149
Flatscreen TV	4,793	166	3.5%	\$300	\$1,445,700
AAA Batteries	20,590	719	3.5%	\$2.99	\$92,741
AA Batteries	20,542	698	3.4%	\$3.84	\$106,118

→ Extreme polarization: Phones achieve 27–30% multi-product rate, while Laptops/Monitors/TVs only 3.5–4.2%. A 7x gap.

5.2. Cross-sell "Dead Zones" — 3 Biggest Missed Opportunities

A. Macbook Pro — "Revenue king, absent from cross-sell"

- Revenue: \$8.04M (23.3% of total) — #1 product
- Multi-product rate: only 4.0% (190/4,720 orders)
- DOES NOT appear in any of the 50 association rules
- When cross-sold: only paired with cables (21x), headphones (20x), AirPods (19x) — all \$12–\$150
- Revenue opportunity: $(4,720 - 190) \times \$220 = \sim \$996,600$ potential. Estimated 10–30% conversion = \$100K–\$300K achievable

B. Flatscreen TV — "Completely invisible"

- Revenue: \$1.45M, 4,793 orders
- Multi-product rate: 3.5% (166 orders)
- ZERO pairs in association rules
- No one buys a TV alongside any other product

C. LG Washer + Dryer — "Most natural combo, zero joint purchases"

- Washer \$600 (666 orders) + Dryer \$600 (646 orders)
- Joint purchases: 0. Not a single order contains both
- Combo \$1,200 could be priced at \$1,099 (save \$101) to incentivize

5.3. Why Do Phones Cross-sell But Laptops/TVs Don't?

Factor	Phones (27–30%)	Laptops/TVs (3–4%)
"Must-have" accessory	Yes — charging cable	No — works out of box
Accessory price	\$12–\$15 (near-zero)	No clear accessory
Functional need	Direct (charge phone)	Not direct

Lift	1.22–1.47	0.03–0.06
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→ Cross-sell only works with a DIRECT FUNCTIONAL LINK. Laptops, TVs, Monitors lack a "must-have accessory" → customers buy standalone.

VI. SUBCATEGORY & CATEGORY CO-OCCURRENCE

6.1. SubCategory Co-occurrence:

Pair	Co-occurrence	Observation
Cables + Phones	2,427	Strongest — Phones need Cables
Headphones + Phones	1,697	Phones + headphones
Cables + Headphones	674	Two accessory types together
Batteries + Headphones	432	Small accessories together
Batteries + Cables	356	Small accessories together

Phones are the "connection hub" — appearing in the 2 strongest pairs, pulling both Cables and Headphones.

SubCategory Co-occurrence

	SubCategory_A	SubCategory_B	Co_Occurrence ▾
1.	Cables	Phones	2,427
2.	Headphones	Phones	1,697
3.	Cables	Headphones	674
4.	Batteries	Headphones	432
5.	Batteries	Cables	356
6.	Headphones	Monitors	248
7.	Cables	Monitors	246

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6.2. Category Co-occurrence:

Pair	Co-occurrence	Observation
Accessories + Electronics	3,236	Strongest — accessories + main

Audio + Electronics	2,073	Headphones + main devices
Accessories + Audio	1,104	Two accessory groups
Home Appliances + any	11–23	Nearly completely independent

Home Appliances (Laundry) almost NEVER purchased alongside other products — highest co-occurrence is only 23.

Category Co-occurrence

		Category_B / Co_Occurrence		
Category_A		Electronics	Audio	Home Appliances
Accessories		3,236	1,104	23
Audio		2,073	-	11
Electronics		-	-	19

VII. KEY CONCLUSIONS & RECOMMENDATIONS

1. PHONE IS THE "ANCHOR KING": Phones are the only product with strong cross-selling power. All cross-sell strategies should begin from phone product pages.

2. ONLY 3 RELATIONSHIPS ARE TRULY MEANINGFUL: Among 50 pairs, only 3 have Lift > 1 — all Phone + Cable. Cross-buying occurs almost exclusively with DIRECT FUNCTIONAL NEED.

3. CROSS-BUYING REVENUE OPPORTUNITY: 4% multi-product orders generate 10.7% of revenue (\$3.70M), AOV \$518 — 2.9x single (\$180). Raising multi rate to 6% adds ~\$1.21M (+3.5%). Macbook triggers \$220 add-on per order.

4. CROSS-SELL "DEAD ZONES": Macbook (\$8.04M, 4.0%), Flatscreen TV (\$1.45M, 3.5%), LG Washer+Dryer (0 joint) — cross-sell only works with functional links, not standalone products.

5. AIRPODS HAS "GONE INDEPENDENT": Lift 0.63 with iPhone — promote as standalone, not tied to iPhone.

6. HOME APPLIANCES SEPARATE CHANNEL: Co-occurrence only 11–23. Washer + Dryer never purchased together — consider \$1,099 combo and dedicated channel.

IMPLEMENTATION RECOMMENDATIONS:

- Tier 1 (deploy now): "Phone + Cable" bundles on product pages — 3 pairs with Lift > 1, proven by data
- Tier 2 (A/B test): "Phone + Headphones" at checkout — volume sufficient but Lift < 1, measure conversion
- LG Combo: Washer + Dryer \$1,099 (save \$101) — currently 0 joint purchases, combo could unlock latent demand